

REAL ANALYSIS

A SELF-TAUGHT APPROACH

Luis Vasquez

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1 Introduction

1.1 Sources

The following notes are taken from the compilation of a few sources

- Zorich, V. A. (2004). Mathematical analysis I (R. Cooke, Trans.). Springer.
- MIT 18.100B Real Analysis, Spring 2025, available at the MIT OCW youtube channel

1.2 Purpose

This notes are taken in a way that is easy to understand math for myself (the author). No obscure proof or incomplete idea will be included, avoiding partial understanding of a certain topic. It also covers the need of having an easy-to-follow approach to *Real Analysis*, making it possible to read this through to revisit known topics without the need of rabbit-hole-ing at other sources. Since I will also be studying the course while taking this notes, the document as a whole will be written by hand without any AI slop nor blind copy/pasting, and since English is not my native language, typos may appear here and there.

*When I say **we**, it means you (the reader) and I. When I say **I/myself**, responsibility only lies on me (the author).*

2 Base knowledge

2.1 What is a Field

“To verify that a set forms a field, we check that multiplication is well-defined. That is, independent of the choice of representative.”

For example, for arbitrary rational numbers Q :

$$\frac{m_1}{n_1} \times \frac{p_1}{q_1}$$

And evaluate an equivalent expression with different representatives of the same numbers:

$$\frac{m_2}{n_2} \times \frac{p_2}{q_2}$$

Given that:

$$\frac{m_1}{n_1} = \frac{m_2}{n_2} \quad \text{and} \quad \frac{p_1}{q_1} = \frac{p_2}{q_2}$$

We wish to show that both products coincide. To review this, we start from the tautology (intuitive truth):

$$\frac{m_1}{n_1} = \frac{m_2}{n_2} \iff m_1 \times n_2 = m_2 \times n_1 \tag{A}$$

$$\frac{p_1}{q_1} = \frac{p_2}{q_2} \iff p_1 \times q_2 = p_2 \times q_1 \tag{B}$$

Then, operating the multiplication using both representatives:

$$\begin{aligned} \frac{m_1}{n_1} \times \frac{p_1}{q_1} &= \frac{m_1 \cdot p_1}{n_1 \cdot q_1} \\ \frac{m_2}{n_2} \times \frac{p_2}{q_2} &= \frac{m_2 \cdot p_2}{n_2 \cdot q_2} \end{aligned}$$

Conveniently, we want to form $m_1 \times n_2$ to use the first ground truth:

$$\begin{aligned} \frac{m_1}{n_1} \times \frac{p_1}{q_1} \times n_2 &= \frac{m_1 \cdot n_2 \cdot p_1}{n_1 \cdot q_1} \\ &= \frac{m_2 \cdot n_1 \cdot p_1}{n_1 \cdot q_1} \quad (\text{replacing using A}) \\ &= \frac{m_2 \cdot p_1}{q_1} \quad (\text{simplifying } n_1) \end{aligned}$$

Applying the same logic for $p_1 \times q_2$ to use the second ground truth:

$$\begin{aligned} \frac{m_1}{n_1} \times \frac{p_1}{q_1} \times n_2 \times q_2 &= \frac{m_2 \cdot p_1 \cdot q_2}{q_1} \\ &= \frac{m_2 \cdot p_2 \cdot q_1}{q_1} \quad (\text{replacing using B}) \\ &= m_2 \cdot p_2 \quad (\text{simplifying } q_1) \end{aligned}$$

Finally, rearranging:

$$\begin{aligned}\frac{m_1}{n_1} \times \frac{p_1}{q_1} \times n_2 \times q_2 &= m_2 \cdot p_2 \\ \frac{m_1}{n_1} \times \frac{p_1}{q_1} &= \frac{m_2 \cdot p_2}{n_2 \cdot q_2} \\ \frac{m_1}{n_1} \times \frac{p_1}{q_1} &= \frac{m_2}{n_2} \times \frac{p_2}{q_2} \quad \square\end{aligned}$$

While not a rigorous proof, this gives us a first step to go from the intuition of a solution (particularly for $\mathbb{Q}$) to a more formal procedure based on the real definition of a field.

2.2 Formal definition

DEFINITION	A field $\mathbb{F}$ is a set with two operations: addition ($\oplus$) and multiplication ($\otimes$), with the following properties:
Field	
2.1	• $x, y \in \mathbb{F} \implies x \oplus y \in \mathbb{F}$ • $x, y \in \mathbb{F} \implies x \oplus y = y \oplus x$ • $x, y, z \in \mathbb{F} \implies (x \oplus y) \oplus z = x \oplus (y \oplus z)$ • $\exists 0 \in \mathbb{F}$ such that $\forall x \in \mathbb{F}, x \oplus 0 = x$ • $\forall x \in \mathbb{F}, \exists (-x) \in \mathbb{F}$ such that $x \oplus (-x) = 0$ • $x, y \in \mathbb{F} \implies x \otimes y \in \mathbb{F}$ • $x, y \in \mathbb{F} \implies x \otimes y = y \otimes x$ • $x, y, z \in \mathbb{F} \implies (x \otimes y) \otimes z = x \otimes (y \otimes z)$ • $\exists 1 \in \mathbb{F}$ such that $\forall x \in \mathbb{F}, x \otimes 1 = x$ • $\forall x \in \mathbb{F} \setminus \{0\}, \exists x^{-1} \in \mathbb{F}$ such that $x \otimes x^{-1} = 1$

The first five properties correspond to the addition operation, and the last five to the multiplication operation. In order to relate both sets of properties, the following axiom is stated:

AXIOM	Let $x, y, z \in \mathbb{F}$. Then
Distributive law	
2.1	$x \otimes (y \oplus z) = x \otimes y \oplus x \otimes z$

THEOREM	For any field $\mathbb{F}$, there exists only one zero element.
Zero uniqueness	
2.1	

PROOF Assume 0_1 and 0_2 are zero elements in $\mathbb{F}$. Then

$$\forall x \in \mathbb{F} : \begin{cases} 0_1 + x = x & \text{(A)} \\ 0_2 + x = x & \text{(B)} \end{cases}$$

Setting $x = 0_2$ in A, and be $x = 0_1$ in B:

$$x = 0_2 \implies \text{(A)} : \quad 0_1 + 0_2 = 0_2 \quad \text{(C)}$$

$$x = 0_1 \implies \text{(B)} : \quad 0_2 + 0_1 = 0_1 \quad \text{(D)}$$

From (C) and commutativity:

$$0_2 + 0_1 = 0_2$$

Comparing this result with (D):

$$\begin{aligned} 0_2 + 0_1 = 0_2 \quad \text{and} \quad 0_2 + 0_1 = 0_1 \\ \implies 0_1 = 0_2 \end{aligned}$$

□

2.3 Order

DEFINITION Ordered sets 2.2	A set S is ordered when it has an ordering “ $<$ ” such that for all $x, y \in S$, exactly one of the following properties holds: 1. $x = y$ 2. $x < y$ 3. $y < x$
DEFINITION Partial Ordering 2.3	A relation “ $<$ ” between pairs of elements of a set X is called a (strict) partial ordering if it follows the properties: 1. $\neg(a < a)$ (irreflexivity); 2. $(a < b) \wedge (b < c) \Rightarrow a < c$ (transitivity); 3. $(a < b) \Rightarrow \neg(b < a)$ (asymmetry).
DEFINITION Total Ordering 2.4	A relation “ $<$ ” between pairs of elements of a set X is called a total ordering (or just <i>Ordering</i>) if it follows the properties: 1. $<$ is a partial ordering; 2. $\forall a, b \in X : a = b \vee a < b \vee b < a$ (Comparability).

DEFINITION
Ordered fields
2.5

A field $\mathbb{F}$ is **ordered** if it is also an ordered set. As a consequence, the following properties apply:

- $x, y \in \mathbb{F}, x < y \implies \forall z \in \mathbb{F}, x \oplus z < y \oplus z$
- $x, y \in \mathbb{F}, 0 < x, y \implies 0 < x \otimes y$

THEOREM
Ordered field
criterion
2.2

Let $\mathbb{F}$ be a field with a well-defined relation > 0 . Suppose that:

- i) for every $x \in \mathbb{F}$, exactly one of $x = 0$, $x > 0$, or $0 > x$ holds;
- ii) if $x, y > 0$, then $x \oplus y > 0$;
- iii) if $x, y > 0$, then $x \otimes y > 0$.

Then defining $x \succ y$ whenever $x \ominus y > 0$ gives an order relation compatible with the field operations, and therefore $\mathbb{F}$ is an ordered field.

PROOF A relation is **well-defined** if its result depends only on the element itself and not on the particular way that element is represented.

Now, we first show that $<$ is an order relation.

For any $x, y \in \mathbb{F}$, trichotomy applied to $y \ominus x$ gives exactly one of

$$y \ominus x = 0, \quad y \ominus x < 0, \quad 0 < y \ominus x.$$

These correspond respectively to

$$x = y, \quad x < y, \quad y < x.$$

Hence the relation satisfies trichotomy.

For transitivity, suppose $x < y$ and $y < z$. Then

$$y \ominus x < 0, \quad z \ominus y < 0.$$

By closure under addition,

$$(z \ominus y) \oplus (y \ominus x) = z \ominus x \\ < 0.$$

Therefore $x < z$, so $<$ is transitive.

Finally, suppose $x < y$. Then $y \ominus x < 0$, and for any $z \in \mathbb{F}$,

$$(y \oplus z) \ominus (x \oplus z) = y \ominus x \\ < 0.$$

Thus $x \oplus z < y \oplus z$.

The multiplication condition is already satisfied by assumption: if $0 < x$ and $0 < y$:

$$x \otimes y > 0.$$

Therefore $<$ is an order relation compatible with the field operations, and hence $\mathbb{F}$ is an ordered field. $\square$

THEOREM 2.3 | Given $\mathbb{F}$ an ordered field. If $x < y$ and $0 < z$, then $x \otimes z < y \otimes z$.

PROOF | We proceed by contradiction. Assume instead that:

$$x \otimes z \geq y \otimes z$$

Then adding $(-x \otimes z)$ on both sides, we maintain the ordering of the expression:

$$\begin{aligned} x \otimes z \oplus (-x \otimes z) &\geq y \otimes z \oplus (-x \otimes z) \\ 0 &\geq y \otimes z \oplus (-x \otimes z) \end{aligned}$$

Now, using Axiom 2.1, we get:

$$0 \geq z \otimes (y \oplus (-x))$$

From the initial conditions, $x < y$ implies $y \oplus (-x) > 0$. Since we also have $z > 0$, we would expect the product of these two to be > 0 by Definition 2.5 (second property). Hence:

$$0 \geq z \otimes (y \oplus (-x)) \quad \text{and} \quad 0 < z \otimes (y \oplus (-x))$$

This is a contradiction. □

From this point forward, for better readability, $+$ and $\cdot$ (or $\times$) will be used instead of $\oplus$ and $\otimes$. They will still represent the abstraction of a field's addition and multiplication operations, without necessarily being the familiar operations we might expect them to be.

2.4 Completeness

DEFINITION | A set $X \subset \mathbb{F}$ ($\mathbb{F}$ ordered field) is said to be *bounded above* (or respectively, *bounded below*) if
 Bounds | $\exists c \in \mathbb{F}$ such that $\forall a \in X, a \leq c$ (or respectively, $c \leq a$). c is called upper (or respectively
 2.6 | lower) bound of X .

DEFINITION | A set that is both bounded above and below, is called *bounded*.
 2.7 |

DEFINITION | An element $a \in X$ is called the *largest* element of X if $\forall x \in X, x \leq a$. Respectively, $a \in X$ is
 2.8 | called the *smallest* element of X if $\forall x \in X, a \leq x$. Simplifying the notation:

$$(a = \max X) := (a \in X \wedge \forall x \in X, x \leq a)$$

$$(a = \min X) := (a \in X \wedge \forall x \in X, a \leq x)$$

These read as *maximal* and *minimal* of X . Now, given this definition, it is important to notice that not every set, not even every bounded set, has a maximal or minimal element. For example:

$$X = \{x \in \mathbb{F} \mid 0 \leq x < 1\}$$

Only has a minimal element (0), but no maximal element, since $1 \notin X$

DEFINITION | The smallest $s \in X \subset \mathbb{F}$ that bounds X from above is called the *least upper bound* of X , and
 Least Upper Bound | denoted $\sup X$ (read "the supremum of X ")
 2.9 |

$$(s = \sup X) := (\forall x \in X, x \leq s) \wedge (\forall s' < s, \exists x' \in X, s' < x')$$

Let us unpack this definition:

- $\forall x \in X$: The following definition applies to every element in X .
- $(x \leq s)$: Given that s is an upper bound for X ...
- $(\forall s' < s) \exists x' \in X (s' < x')$:
 - $(\forall s' < s)$: Considering any arbitrary s' smaller than our upper bound s
 - $\exists x' \in X$: There will be an element x' in X , so that...
 - $(s' < x')$: It is larger than the s' , showing that s' is not an upper bound.

So in summary, s is $\sup X$ if and only if, s is an upper bound, and no smaller number s' is an upper bound of X , because we can find an x' that is not bounded by it.

DEFINITION
Greatest Lower
Bound
2.10

Similarly, the greatest $i \in X \subset \mathbb{F}$ that bounds X below is called the *greatest lower bound of X* , and denoted $\inf X$ (read "the infimum of X ")

$$(i = \inf X) := (\forall x \in X, i \leq x) \wedge (\forall i' > i, \exists x' \in X, x' < i')$$

Thus, we have now the following properties from the definitions:

$$\sup X := \min\{c \in \mathbb{F} \mid \forall x \in X (x \leq c)\}$$

$$\inf X := \max\{c \in \mathbb{F} \mid \forall x \in X (c \leq x)\}$$

It is important to note that the supremum and infimum of a set, as defined above, may not exist in an arbitrary ordered field $\mathbb{F}$. The definitions above specify what $\sup X$ and $\inf X$ mean *if they exist*, but they do not guarantee existence. We will address this issue shortly.

THEOREM
Characterization of
the Supremum
2.4

Let $X \subset \mathbb{F}$ be a nonempty set in an ordered field $\mathbb{F}$, and let $s \in \mathbb{F}$. Then $s = \sup X$ if and only if

$$\forall x \in X : x \leq s \wedge \forall \epsilon > 0, \exists x \in X : s - \epsilon < x.$$

PROOF Suppose $s = \sup X$. By definition, s is an upper bound of X :

$$\forall x \in X : x \leq s.$$

Also, by the definition of the least upper bound, no $s' < s$ is an upper bound of X . Therefore:

$$\forall s' < s, \exists x \in X : s' < x.$$

Let $\epsilon > 0$ and take $s' = s - \epsilon$. Since $\epsilon > 0$:

$$s - \epsilon < s.$$

Therefore:

$$\exists x \in X : s - \epsilon < x.$$

Conversely, suppose that:

$$\forall x \in X : x \leq s \wedge \forall \epsilon > 0, \exists x \in X : s - \epsilon < x.$$

The first condition shows that s is an upper bound of X .

Now, let $s' < s$. Then:

$$\epsilon := s - s' > 0.$$

By assumption:

$$\exists x \in X : s - \epsilon < x.$$

Since $\epsilon = s - s'$:

$$s' < x.$$

Therefore, s' is not an upper bound of X . Hence, no number smaller than s is an upper bound of X , so s is the least upper bound of X .

Therefore:

$$s = \sup X.$$

□

THEOREM
Uniqueness of
Supremum
2.5

Let $X \subset \mathbb{F}$ be a nonempty set in an ordered field $\mathbb{F}$. If X has a supremum, then this supremum is unique.

PROOF

Suppose s_1 and s_2 are both suprema of X . We will show that $s_1 = s_2$.

Since $s_1 = \sup X$, we know that s_1 is an upper bound of X . Since $s_2 = \sup X$, we know that s_2 is the *least* upper bound of X . Therefore:

$$s_2 \leq s_1$$

By the same reasoning (swapping the roles of s_1 and s_2), since s_2 is an upper bound and s_1 is the least upper bound:

$$s_1 \leq s_2$$

By the antisymmetry property of order in $\mathbb{F}$, we have $s_1 \leq s_2$ and $s_2 \leq s_1$, which implies:

$$s_1 = s_2$$

Therefore, the supremum is unique. □

The proof for the uniqueness of the infimum is analogous.

EXAMPLE
2.1

Consider the ordered field $\mathbb{Q}$ of rational numbers, and let:

$$S = \{x \in \mathbb{Q} \mid x^2 < 2\}$$

The set S is nonempty (for instance, $1 \in S$) and bounded above (for instance, $2 \in \mathbb{F}$ is an upper bound). However, S has no supremum in $\mathbb{Q}$: if such a supremum existed, it would equal $\sqrt{2} \notin \mathbb{Q}$, so S has no least upper bound within the rational numbers.

This shows that not every ordered field has the property that bounded sets possess suprema.

DEFINITION Complete Ordered Field 2.11	An ordered field $\mathbb{F}$ is called <i>complete</i> if every nonempty subset of $\mathbb{F}$ that is bounded above has a supremum in $\mathbb{F}$.
--	---

PROPOSITION 2.1	If $\mathbb{F}$ is a complete ordered field, then every nonempty subset of $\mathbb{F}$ that is bounded below has an infimum in $\mathbb{F}$.
---------------------------	--

PROOF Let $X \subset \mathbb{F}$ be nonempty and bounded below. Define:

$$X' = \{-x \mid x \in X\}$$

1. X' is nonempty and bounded above

Since X is nonempty, $\exists x_0 \in X \Rightarrow -x_0 \in X'$ by definition of X' . Thus $X' \neq \emptyset$.
Now, X is bounded below by the proposition, so there exists $c \in \mathbb{F}$ such that:

$$\begin{aligned} \forall x \in X, \quad c &\leq x \\ \forall x \in X, \quad -x &\leq -c \end{aligned}$$

Since $-x \in X'$ and $-c \in \mathbb{F}$, we can be sure that X' is bounded above in $\mathbb{F}$.

Considering $\mathbb{F}$ is complete, and we proved X' is nonempty and bounded above, X' has a supremum in $\mathbb{F}$.

2. $-s$ is a lower bound for X .

Let $s = \sup X'$ and $x \in X$ be arbitrary. Then $-x \in X'$ by definition of X' . Since s is an upper bound for X' :

$$\begin{aligned} \forall -x \in X', \quad -x &\leq s \\ \Rightarrow \quad -s &\leq x \end{aligned}$$

Showing that $-s$ is a lower bound for X .

3: $-s$ is the greatest lower bound for X .

Let $\ell \in \mathbb{F}$ be any lower bound for X .

Since ℓ is a lower bound for X :

$$\begin{aligned} \forall x \in X, \quad \ell &\leq x \\ \Rightarrow \quad -x &\leq -\ell \end{aligned}$$

Meaning that $-\ell$ is an upper bound for any $-x \in X'$. Since $s = \sup X'$, we have:

$$\begin{aligned} s &\leq -\ell \\ \Rightarrow \ell &\leq -s \end{aligned}$$

Showing that $-s$ is greater than or equal to every lower bound of X . Therefore, $-s = \inf X$. $\square$

THEOREM
Least Upper Bound
Principle
2.6

Let $\mathbb{F}$ be a complete ordered field. Every nonempty subset of $\mathbb{F}$ that is bounded above has a unique least upper bound in $\mathbb{F}$.

PROOF

Let $X \subset \mathbb{F}$ be nonempty and bounded above.

Existence: Since $\mathbb{F}$ is complete, by definition of completeness, X has a supremum in $\mathbb{F}$.

Uniqueness: By the Uniqueness of Supremum theorem, this supremum is unique. Therefore, X has a unique least upper bound. $\square$

2.5 Archimedean property

Now that we achieved completeness on an ordered field, we can now extend the definitions and properties into a theorem/property that uses the result from the *Least Upper Bound principle* (Theorem 2.6).

THEOREM
2.7

Let $\mathbb{F}$ be a complete ordered field, then

$$\forall x \in \mathbb{F}, \exists n \in \mathbb{N} \subset \mathbb{F} : n > x$$

PROOF

We first define the subset $\mathbb{N} \subset \mathbb{F}$ **inductively**, to work with a generalized version of the set of “natural numbers”

1. $1_{\mathbb{F}}$ is the multiplicative identity in $\mathbb{F}$
2. $n + 1_{\mathbb{F}} \in \mathbb{N}, \forall n \in \mathbb{N}$ defines the induction step over $\mathbb{N}$, using the addition operation from the field $\mathbb{F}$

We proceed by contradiction. Suppose $\exists x \in \mathbb{F}, \forall n \in \mathbb{N} :$

$$n \leq x$$

Meaning that $\mathbb{N}$ is bounded above. Now, since $\mathbb{N}$ is not empty, and is bounded by an element of the (complete ordered) field that contains it ($x \in \mathbb{F}$), then by *L.U.B. principle* (Theorem 2.6), $\mathbb{N}$ must have a supremum.

Let $s := \sup \mathbb{N} \in \mathbb{F}$ be that supremum of $\mathbb{N}$. Then, since s is the least upper bound, any element that is less than s will no longer be an upper bound for $\mathbb{N}$. Conveniently we take $s - 1_{\mathbb{F}}$:

$$\begin{aligned} \exists n_0 \in \mathbb{N} : s - 1_{\mathbb{F}} &< n_0 \\ \Rightarrow s &< n_0 + 1_{\mathbb{F}} \end{aligned}$$

But by our inductive definition of $\mathbb{N} : (n_0 + 1_{\mathbb{F}}) \in \mathbb{N}$. So we found out that s ($\sup \mathbb{N}$) fails to be an upper bound for an element in $\mathbb{N}$.

Therefore, our original assumption of x must be wrong. $\square$

Now, it is important to notice that $\mathbb{F}$ implies an Archimedean field, but the opposite won't necessarily be true.

EXAMPLE
2.2

Using an Archimedean field, that is not complete, let us take $\mathbb{F} = \mathbb{Q}$

$$x \in \mathbb{Q} \wedge n \in \mathbb{N} \subset \mathbb{Q}$$

It is true that, for any x there is a natural number n so that

$$n > x$$

But invoking the previous example (Example 2.1) we can take a subset of $\mathbb{Q}$ that is not bounded in $\mathbb{Q}$, failing to properly form a complete set.

Closing this chapter, let us emphasize the fact that the construction of $\mathbb{N} \subset \mathbb{F}$ by induction was crucial to generalize the property for any complete ordered field. Next we will explain why this wasn't necessary at all.

2.6 Problems & Exercises

Problem 2.1. If $A \subset B \subset \mathbb{R} \Rightarrow \sup A \leq \sup B \wedge \inf A \geq \inf B$

SOLUTION.

i) Since $\sup B$ is an upper bound of $B : \forall b \in B : b \leq \sup B$

Since $A \subset B, \forall a \in A$ also belongs to $B \rightarrow \forall a \in A : a \leq \sup B$.

Therefore, $\sup B$: upper bound of A . Finally, since $\sup A \in A \rightarrow \sup A \leq \sup B$ as well.

$$\Rightarrow \sup A \leq \sup B.$$

ii) Since $\inf B$ is a lower bound of $B : \forall b \in B : \inf B \leq b$

Since $A \subset B, \forall a \in A$ also belongs to $B \rightarrow \forall a \in A : \inf B \leq a$.

Therefore, $\inf B$: lower bound of A . Finally, since $\inf A \in A \rightarrow \inf B \leq \inf A$ as well.

$$\Rightarrow \inf A \geq \inf B.$$

■

Problem 2.2. Let $X, Y \subset \mathbb{R}$ be non empty sets, such that $x \leq y, \forall x \in X, \forall y \in Y$

Then, proof the following hold: X is bounded above, Y is bounded below, and $\sup X \leq \inf Y$

SOLUTION.

i) Since Y is non-empty, $\exists y \in Y$. Then $\forall x \in X : x \leq y \rightarrow y$ is an upper bound of X , so X is bounded above.

ii) Since X is non-empty, $\exists x \in X$. Similarly $\forall y \in Y : x \leq y \rightarrow x$ is a lower bound of Y , so Y is bounded below.

iii) Since Y is bounded below and X is bounded above: $\exists \inf Y \wedge \sup X$ so that:

$$\begin{aligned} \forall x \in X, \forall y \in Y : x \leq y &\rightarrow \forall x \in X : (\forall y \in Y : x \leq y) \\ &\rightarrow \forall x \in X : x \text{ is a lower bound of } Y \\ &\rightarrow \forall x \in X : x \leq \inf Y \end{aligned}$$

Thus, $\inf Y$ is an upper bound of X .

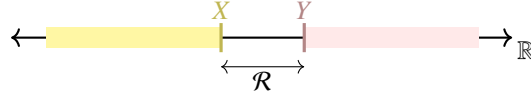
By the definition of $\sup X$ being the least upper bound, any other upper bound will be greater:

$$\Rightarrow \sup X \leq \inf Y.$$

■

Problem 2.3. Let $X, Y \subset \mathbb{R}$ be non empty sets, such that $x \leq y$, $\forall x \in X, \forall y \in Y$, and suppose that $X \cup Y = \mathbb{R}$. Prove that $\sup X = \inf Y$.

SOLUTION. From the previous result, we have: $\sup X \leq \inf Y$. Now, let's assume that $\sup X < \inf Y$ strictly, and try to reach the absurd. Intuition tells me that the idea is to expect items between X and Y , and to prove that this would violate the initial conditions.



Assuming $\sup X < \inf Y$:

$$\begin{aligned}
 & \exists r \in \mathbb{R} : \sup X < r < \inf Y \\
 \rightarrow & \mathcal{R} := \{r \in \mathbb{R} / \sup X < r < \inf Y\} \neq \emptyset \\
 & \rightarrow r \notin X \wedge r \notin Y \\
 & \rightarrow r \notin X \cup Y \\
 & \rightarrow r \notin \mathbb{R} \quad (\rightarrow \leftarrow)
 \end{aligned}$$

$$\therefore \sup X = \inf Y.$$

■

Problem 2.4. *Dedekind's Theorem:* Let $X, Y \subset \mathbb{R}$ be non empty sets, such that $x \leq y$, $\forall x \in X, \forall y \in Y$, and suppose that $X \cup Y = \mathbb{R}$. Then either X has a maximal element, or Y has a minimal element.

SOLUTION. Let $s := \sup X = \inf Y$ (by the previous problem), and since $X \cup Y = \mathbb{R}$, and $s \in \mathbb{R}$, we must have $s \in X$ or $s \in Y$.

Case 1: $s \in X$.

Since $s = \sup X$ is an upper bound of X , and $s \in X$:

$$\forall x \in X, x \leq s \quad \wedge \quad s \in X \Rightarrow s = \max X$$

Case 2: $s \in Y$.

Since $s = \inf Y$ is a lower bound of Y , and $s \in Y$:

$$\forall y \in Y, s \leq y \quad \wedge \quad s \in Y \Rightarrow s = \min Y$$

$\therefore X$ has a maximal element, or Y has a minimal element.

■

Problem 2.5. Let $A, B \subset \mathbb{R}$, and sets defined:

- $A + B = \{a + b, a \in A, b \in B\}$
- $A.B = \{a.b, a \in A, b \in B\}$

Determine if the following is true:

- i) $\sup(A + B) = \sup A + \sup B$
- ii) $\sup(A.B) = \sup A . \sup B$

SOLUTION.

- i) $\forall a \in A, b \in B \rightarrow a \leq \sup A \wedge b \leq \sup B$

Adding up both inequalities:

$$\begin{aligned} a + b &\leq \sup A + \sup B, \forall a \in A, b \in B \\ \Rightarrow \sup A + \sup B &\text{ is upper bound of } A + B \end{aligned}$$

To show that $\sup A + \sup B$ is the least upper bound of $A + B$, it is sufficient to show that:

$$\forall \epsilon > 0, \exists a + b \in A + B : \sup A + \sup B - \epsilon < a + b \leq \sup A + \sup B.$$

Let $\epsilon > 0$ be arbitrary. Since $\epsilon/2 > 0$, by the characterization of $\sup A$ (Theorem 2.4):

$$\exists a \in A : \sup A - \frac{\epsilon}{2} < a \leq \sup A.$$

Similarly, by the characterization of $\sup B$:

$$\exists b \in B : \sup B - \frac{\epsilon}{2} < b \leq \sup B.$$

Adding both inequalities:

$$\begin{aligned} \sup A + \sup B - \frac{\epsilon}{2} - \frac{\epsilon}{2} &< a + b \leq \sup A + \sup B \\ \Rightarrow \sup A + \sup B - \epsilon &< a + b \leq \sup A + \sup B \\ \Rightarrow \sup A + \sup B &: \text{least upper bound of } A + B \\ \therefore \sup(A + B) &= \sup A + \sup B. \end{aligned}$$

- ii) By counter-example, let $A = \{1, 2\}$ and $B = \{-3, -4\}$. Then:

$$\begin{aligned} A.B &= \{-3, -4, -6, -8\} \\ \sup(A.B) &= -3 \neq (2. -3) = \sup A . \sup B. (\rightarrow \leftarrow) \end{aligned}$$

■

Problem 2.6. Let $-A$ be the set of numbers of the form $-a$, where $a \in A \subset \mathbb{R}$.

Show that $\sup(-A) = -\inf A$.

SOLUTION. Let $s := \sup(-A)$, so that $\forall -a \in -A : -a \leq s$.

$$\begin{aligned} \forall -a \in -A : -a &\leq s \\ \rightarrow -(-a) &\geq -s \\ \rightarrow -s &\leq a \\ \rightarrow -s &: \text{lower bound of } A \end{aligned}$$

Additionally, for $-A$ we have by the supremum's characterization:

$$\begin{aligned} \forall \epsilon > 0, \exists x \in -A : s - \epsilon &< x \leq s \\ (\text{mul. } -1) : -s + \epsilon &> -x \geq -s, -x \in A \\ (\text{fixing ineq. side}) : -s &\leq -x < -s + \epsilon \\ \Rightarrow -s &: \text{greatest lower bound of } A \end{aligned}$$

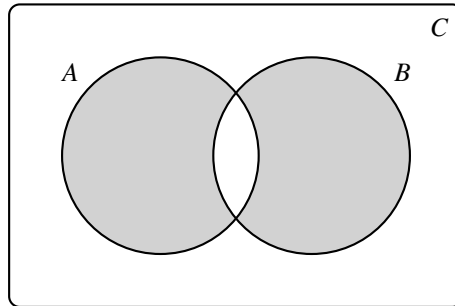
Finally:

$$\begin{aligned} -s &= \inf(A) \\ \therefore \sup(-A) &= -\inf(A) \end{aligned}$$

■

Problem 2.7. Let A , B , and C be sets such that $A \subset C$, $B \subset C$, $A \setminus B \neq \emptyset$, and $B \setminus A \neq \emptyset$. We introduce a partial ordering into this triple of sets. Exhibit the maximal and minimal elements of the set $\{A, B, C\}$.

SOLUTION. This is the setup that the problem proposes:



Where $A \setminus B \wedge B \setminus A$ are not empty. Then for set $\{A, B, C\}$:

$$\begin{aligned} A \subset C \wedge B \subset C &\rightarrow A \subset B \subset C \\ &\rightarrow \max_C \{A, B, C\} = C \end{aligned}$$

Now, since $A \not\subset B \wedge B \not\subset A$:

$$\rightarrow \min_C \{A, B, C\} \text{ is not unique.}$$

Both A and B are minimal elements (sets) but none of them is the *least*. ■

Problem 2.8. Show that, just like the set $\mathbb{Q}$ of rational numbers, the set $\mathbb{Q}(\sqrt{n})$ of numbers of the form $a + b\sqrt{n}$, where $a, b \in \mathbb{Q}$ and n is a fixed natural number that is not the square of any integer, is an ordered set satisfying the principle of Archimedes but not the axiom of completeness.

SOLUTION. Considering that $a, b \in \mathbb{Q}$ and that $\mathbb{Q}$ is an Archimedean (Ordered Field), we can conveniently say:

$$\exists N_a, N_b \in \mathbb{N} : |a| < N_a \wedge |b| < N_b$$

And since $n \in \mathbb{N}$ is a fixed number, it also satisfies:

$$\sqrt{n} < n + 1$$

Correctly adding and multiplying these three inequalities:

$$\begin{aligned} |a| &< N_a \\ |b| &< N_b \\ \sqrt{n} &< n + 1 \\ \rightarrow |a| + |b|\sqrt{n} &< N_a + N_b(n + 1) \end{aligned}$$

On the left side, we can consider $|a| + |b|\sqrt{n} = |a + b\sqrt{n}|$. Then, using the triangle inequality:

$$|a + b\sqrt{n}| \leq |a| + |b|\sqrt{n} = |a| + |b|\sqrt{n} \rightarrow |a + b\sqrt{n}| < N_a + N_b(n + 1)$$

Finally, by simple absolute value property

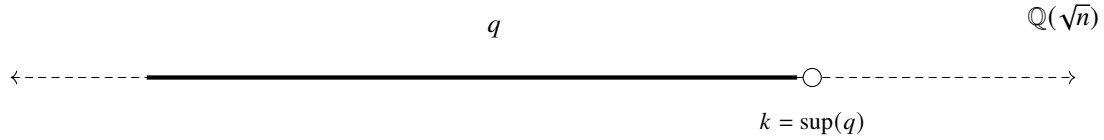
$$a + b\sqrt{n} \leq |a + b\sqrt{n}|$$

Now, let $x = a + b\sqrt{n} \in \mathbb{Q}(\sqrt{n})$, and $N = N_a + N_b(n + 1) \in \mathbb{N}$:

$$\begin{aligned} \Rightarrow x &< N, x \in \mathbb{Q}(\sqrt{n}), N \in \mathbb{N} \\ \therefore \mathbb{Q}(\sqrt{n}) &: \text{Archimedean.} \end{aligned}$$

Now, to prove that this is not a complete field, we need to form a subset $q \subset \mathbb{Q}(\sqrt{n})$ that somehow has its supremum in $\mathbb{R}$, but outside of $\mathbb{Q}(\sqrt{n})$. I found out myself that this could go one of two ways: the *skadoodle-out-of-thin-air* answer, or the *intuitively-looking-for-an-answer* answer. Of course I will do the second one, but in case this is too ugly for the sake of an arbitrary exercise, please feel free to skip (by the way, answer is $\sqrt[3]{n}$).

Now to build q we could start with this rough idea:



Let's first assume there exists a k outside of $\mathbb{Q}(\sqrt{n})$, so that our chosen set q is:

$$q = \{x \in \mathbb{Q}(\sqrt{n}), x < k\}$$

By doing this, we guarantee that k bounds q . Again, we are trying to prove that this is the supremum in such a way that we guarantee its existence outside of $\mathbb{Q}(\sqrt{n})$.

To prove that this is the lowest upper bound, let's assume the "real" supremum is actually a k' :

i) Assume $\exists k' : k' < k$. By density in $\mathbb{Q}$:

$$\begin{aligned} \exists x \in \mathbb{Q} : k' < x < k \\ \text{Since } \mathbb{Q} \subset \mathbb{Q}(\sqrt{n}) &\rightarrow \exists x \in \mathbb{Q}(\sqrt{n}) : k' < x < k. \\ &\rightarrow x \in q \\ &\rightarrow k' \text{ not L.U.B of } q \end{aligned}$$

And so, by showing that every number smaller than k fails to be an upper bound for q , while k itself is an upper bound, we prove that

$$k = \sup(q).$$

Now, this is just half of the proof. At the beginning we assumed that this k exists, and I don't appreciate left over assumptions. The next proof is to make sure that there is at least a number that allows us to create q . Let's call it λ and try to prove $\exists \lambda \in \mathbb{R} : \lambda \notin \mathbb{Q}(\sqrt{n})$.

$$\lambda \notin \mathbb{Q}(\sqrt{n}) \leftrightarrow \lambda \neq a + b\sqrt{n}$$

My intent is to apply some sort of transformation to both sides, and get to an absurd. I tried several things (log, factorial, exp, etc), but the only clue I got back is that the $\sqrt{n}$ was in the way. Then I realized that, as long as I can find a λ whose square is of the form $A + B\sqrt{n}$, I could compare coefficients and try to force an absurd.

In other words, I want

$$\lambda^2 = A + B\sqrt{n}.$$

The simplest choice seems to be

$$A = 0, \quad B = 1,$$

which gives

$$\begin{aligned}\lambda^2 &= \sqrt{n} \\ \lambda &= \sqrt{\sqrt{n}} \\ \lambda &= \sqrt[4]{n}.\end{aligned}$$

Now, we still need to prove that this choice of λ is actually outside of $\mathbb{Q}(\sqrt{n})$. So, let's assume the opposite and try to force an absurd:

$$\sqrt[4]{n} = a + b\sqrt{n}$$

for some $a, b \in \mathbb{Q}$.

Trying to square everything:

$$\begin{aligned}\sqrt{n} &= (a + b\sqrt{n})^2 \\ &= a^2 + b^2n + 2ab\sqrt{n} \\ &= (a^2 + b^2n) + (2ab)\sqrt{n}.\end{aligned}$$

Now, the trick is to realize that both sides are written in the form $A + B\sqrt{n}$, so we can compare coefficients. Since $\sqrt{n} \notin \mathbb{Q}$, this representation is unique. Therefore:

$$\begin{aligned}0 &= a^2 + b^2n \\ 1 &= 2ab.\end{aligned}$$

Now, $a^2 + b^2n \geq 0$ for all $a, b \in \mathbb{Q}$ and $n \in \mathbb{N}$. For it to be zero, we must have

$$a = b = 0.$$

But then

$$2ab = 0 \neq 1,$$

which is a contradiction.

Therefore,

$$\sqrt[4]{n} \notin \mathbb{Q}(\sqrt{n}).$$

Finally, we can define

$$q = \{x \in \mathbb{Q}(\sqrt{n}), x < \sqrt[4]{n}\}.$$

I have the hunch that it is possible to find other numbers besides $\sqrt[4]{n}$ that also satisfy to be an outsider supremum. I raised the same comparison for λ to the third and fourth exponent, and could only find an absurd scenario for the latter one, making me think that these numbers live in a set like

$$\Lambda = \{\sqrt[2k]{n}, k \in \mathbb{N}\}.$$

It would need induction to be proven, but I don't have that kind of time. ■

Problem 2.9. Show that the set $\mathbb{Q}(x)$ of rational fractions

$$\frac{a_0 + a_1x + \cdots + a_mx^m}{b_0 + b_1x + \cdots + b_nx^n}$$

with coefficients in $\mathbb{Q}$ or $\mathbb{R}$ becomes an ordered field, but not an Archimedean ordered field, when the order relation $R_{m,n} \succ 0$ is defined to mean $a_mb_n > 0$ and the usual arithmetic operations are introduced.

SOLUTION. First let's address the order defined as:

$$R_{m,n} \succ 0 \leftrightarrow a_mb_n > 0$$

To prove order, we must check the set's behavior over trichotomy, and closure over addition and multiplication (Theorem 2.2). We must also make sure that the order is well-defined, since the same rational function can have more than one representation.

i) **Well-definedness:**

Suppose that the same rational function can be written in two different ways:

$$\frac{P(x)}{Q(x)} = \frac{R(x)}{S(x)}.$$

Then

$$P(x)S(x) = R(x)Q(x).$$

Let the leading coefficients of P, Q, R, S be respectively a_m, b_n, c_p, d_q . Comparing the leading coefficients on both sides gives:

$$a_md_q = c_pb_n.$$

Therefore, the two representations have the same sign under the proposed order. Hence, whether $R_{m,n} \succ 0$ or $R_{m,n} \prec 0$ does not depend on the particular representation chosen.

ii) **Trichotomy:**

Since the given order “ $\succ$ ” subtly misses defining the equality, and we need to check if a given element could be either greater, less than or **equal** to zero, we need to consider the case where the numerator is the zero polynomial.

If the numerator is zero, then:

$$R_{m,n} = 0.$$

Otherwise, the leading coefficients a_m, b_n are both non-zero, and therefore exactly one of $a_mb_n > 0$ or $a_mb_n < 0$ holds. We have:

$$\begin{aligned} a_mb_n > 0 &\leftrightarrow R_{m,n} \succ 0 \\ a_mb_n < 0 &\leftrightarrow -a_mb_n > 0 \leftrightarrow -R_{m,n} \succ 0 \leftrightarrow 0 \succ R_{m,n}. \end{aligned}$$

Therefore, exactly one of

$$R_{m,n} = 0, \quad R_{m,n} \succ 0, \quad 0 \succ R_{m,n}$$

holds, satisfying trichotomy.

iii) **Closure over $\oplus$:**

Solving for any pair of $R_{m,n} \oplus R_{p,q}$, $R_{m,n}, R_{p,q} \succ 0$:

$$\begin{aligned} R_{m,n} \oplus R_{p,q} &= \frac{a_0 + \cdots + a_m x^m}{b_0 + \cdots + b_n x^n} + \frac{c_0 + \cdots + c_p x^p}{d_0 + \cdots + d_q x^q} \\ &= \frac{(a_m d_q x^{m+q}) + (c_p b_n x^{p+n}) + \mathbb{S}_1}{b_n d_q x^{n+q} + \mathbb{S}_2}. \end{aligned}$$

We will just consider the terms containing the leading degree of both sides of the fraction. Since the order just depends on these, we can skip evaluating the rest by indicating a left over expression added up as $\mathbb{S}_{1,2}$. Then, we have two possibilities for the coefficient of the leading degree: $a_m d_q$ or $c_p b_n$.

If the degrees $m+q$ and $p+n$ are different, the term with the larger degree leads. If they are equal, the two leading terms are added together. In this case, we need to make sure that they do not cancel.

Noticing as well that since both $R_{m,n}, R_{p,q} \succ 0 \rightarrow a_m b_n > 0 \wedge c_p d_q > 0$, we evaluate:

- If $a_m d_q$ leads:

$$a_m d_q b_n d_q = a_m b_n d_q^2 > 0.$$

- If $c_p b_n$ leads:

$$c_p b_n b_n d_q = c_p d_q b_n^2 > 0.$$

Therefore, in either case, the leading coefficient of the numerator has the same sign as the leading coefficient of the denominator. If $m+q = p+n$, then both leading coefficients have the same sign, so they cannot cancel and their sum preserves that sign.

Meaning that either way, we preserve the order on addition:

$$\Rightarrow R_{m,n} \oplus R_{p,q} \succ 0.$$

iv) **Closure over $\otimes$:**

$$\begin{aligned} R_{m,n} \otimes R_{p,q} &= \frac{a_0 + \cdots + a_m x^m}{b_0 + \cdots + b_n x^n} \cdot \frac{c_0 + \cdots + c_p x^p}{d_0 + \cdots + d_q x^q} \\ &= \frac{(a_m c_p x^{m+p}) + \mathbb{S}_1}{b_n d_q x^{n+q} + \mathbb{S}_2}. \end{aligned}$$

Under the same condition $a_m b_n > 0 \wedge c_p d_q > 0$, we get that

$$a_m c_p b_n d_q = (a_m b_n)(c_p d_q) > 0.$$

Therefore:

$$\Rightarrow R_{m,n} \otimes R_{p,q} \succ 0.$$

The usual arithmetic operations on rational functions give the usual field structure, and the properties above show that the proposed order is compatible with addition and multiplication. Therefore, $\mathbb{Q}(x)$ is an ordered field.

Now, to prove that $\mathbb{Q}(x)$ is not an Archimedean ordered field, we need to prove that there is at least one element in $\mathbb{Q}(x)$ that is larger than every element of $\mathbb{N}$.

Let $k \in \mathbb{N}$. Consider $x - k \in \mathbb{Q}(x)$:

$$x - k = \frac{x - k}{1}.$$

The leading coefficient of the numerator is 1, and the leading coefficient of the denominator is also 1, so:

$$1 \cdot 1 > 0 \Rightarrow x - k \succ 0.$$

Therefore:

$$\begin{aligned} x - k \succ 0 &\Rightarrow x \succ k, & \forall k \in \mathbb{N} \\ &\Rightarrow \exists x \in \mathbb{Q}(x) : x > k, & \forall k \in \mathbb{N} \\ &\therefore \mathbb{Q}(x) \text{ is not Archimedean.} \end{aligned}$$

■

3 $\mathbb{R}$ as the complete ordered field

3.1 $\mathbb{F}$ is $\mathbb{R}$

Turns out there is a reason why it is called **Real** Analysis. So far we have worked over a generic *Complete Ordered Field* $\mathbb{F}$, that was also an *Archimedean* following the expected properties. We have finally reached the point where we can state that **any** set that manages to follow all these properties is actually identical to the set of real numbers $\mathbb{R}$. In fact, $\mathbb{R}$ is the only set that accomplishes this.

Most *Real Analysis* courses won't go into proving this, but as I already mentioned in the first chapter, I want to cover all doubts and missing points existent during a course like this.

We will go through the initial proofs needed to understand the main one. At the beginning it's possible that some of them look unnecessary or trivial, but I don't want to assume that much. Better to be safe than sorry.

Our starting point will be defining what it means to have "equal fields".

DEFINITION Field homomorphism 3.1	Let F, F' fields. A function $\phi : F \rightarrow F'$ is a homomorphism between F and F' iff. it preserves the defined operations in both fields: $\phi(x + y) = \phi(x) + \phi(y), \forall x, y \in F$ $\phi(x \cdot y) = \phi(x) \cdot \phi(y), \forall x, y \in F$
---	--

DEFINITION Field isomorphism 3.2	Let F, F' fields. A function $\phi : F \rightarrow F'$ is an isomorphism iff. it is a homomorphism and also a bijection
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LEMMA Injectivity by order preservation 3.1	Let F, F' fields, $\phi : F \rightarrow F'$. If ϕ preserves order from $\mathbb{F}$: $x < y \Rightarrow \phi(x) < \phi(y), x, y \in F$ Then ϕ is injective.
---	---

PROOF	To establish injectivity, we must show that for any two identical images of the function ϕ, we get that their arguments were the same, guaranteeing the uniqueness of the image: Let $x, y \in \mathbb{F}$ so that $\phi(x) = \phi(y)$ By trichotomy of F: Case 1: $x < y$: $\rightarrow \phi(x) < \phi(y) (\rightarrow \leftarrow)$ Case 2: $y < x$: $\rightarrow \phi(y) < \phi(x) (\rightarrow \leftarrow)$ $\Rightarrow x = y$ Then, ϕ is injective. □
--------------	---

LEMMA Surjectivity by construction 3.2	Let F, F' fields, $\phi : F \rightarrow F'$. Define: $\text{Im}_F := \{\phi(x), x \in F\}$ Then, the bounded function $\phi' : F \rightarrow \text{Im}_F$ is surjective
--	--

PROOF	To prove surjectivity, we need to show that every element in $y \in \text{Im}_F$ has an $x \in F$ such that $\phi'(x) = y$. Using the definition of our constructed set: $\begin{aligned} \forall y \in \text{Im}_F &\Leftrightarrow y \in \{\phi(x), x \in F\} \\ &\Rightarrow \exists x \in F / \forall y \in \text{Im}_F : y = \phi'(x) \end{aligned}$ Finally proving that for any image of $\phi' \in \text{Im}_F$ we will have a preimage $x \in F$. Then ϕ' is surjective. □
--------------	---

On this first part, this surjectiveness by construction might be the bit that doesn't quite seem correct. Isn't it trivial to have surjectiveness if we grab the method ϕ and limit it to its image ϕ' ?

The key here is that we first defined the set of images Im_F , and *then* proved the surjectiveness in it. The lemma confirms that the set we defined was *just right*. Not so big to leave some elements unreachable, and not too small to make undefined preimages at some spots.

3.2 Sets construction

Now, to make it reasonable for a complete ordered field $\mathbb{F}$ to be isomorphic to our known $\mathbb{R}$, we need to start shaping it using the parts that make it up. Think of $\mathbb{R}$ as the joint set of irrational and rational numbers, which contain the other types of number as well.

From this idea, we need to build $\mathbb{N}_{\mathbb{F}}$, $\mathbb{Z}_{\mathbb{F}}$, $\mathbb{Q}_{\mathbb{F}}$ and $\mathbb{I}_{\mathbb{F}}$ to finally get this $\mathbb{F}$ we want. This will all make sense in the last part of the process, and it is true that it will be a bit of a hustle to construct all of it (specially $\mathbb{I}_{\mathbb{F}}$ from $\mathbb{Q}_{\mathbb{F}}$), but we need

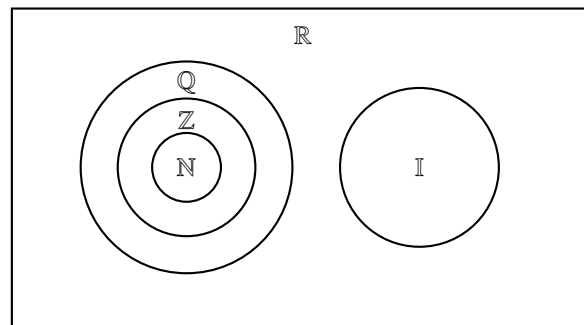


Figure 1: Hierarchy of number sets: Natural numbers ($\mathbb{N}$), Integers ($\mathbb{Z}$), Rationals ($\mathbb{Q}$), Irrationals ($\mathbb{I}$), and Real numbers ($\mathbb{R}$).

to go step by step.

DEFINITION
 $\mathbb{N}_{\mathbb{F}}$ by induction
 3.3

Let $\mathbb{F}$ be *the* complete ordered field. Using $1_{\mathbb{F}}$: multiplicative neutral of $\mathbb{F}$, we declare the following element construction:

$$\begin{aligned} 1_{\mathbb{F}} &:= 1_{\mathbb{F}} \\ 2_{\mathbb{F}} &:= 1_{\mathbb{F}} + 1_{\mathbb{F}} \\ 3_{\mathbb{F}} &:= 1_{\mathbb{F}} + 1_{\mathbb{F}} + 1_{\mathbb{F}} \\ &\vdots \\ n_{\mathbb{F}} &:= \sum_1^n 1_{\mathbb{F}} \end{aligned}$$

Finally, we define the set:

$$\mathbb{N}_{\mathbb{F}} := \{1_{\mathbb{F}}, 2_{\mathbb{F}}, 3_{\mathbb{F}}, \dots\} = \{n_{\mathbb{F}} : n \in \mathbb{N}\}$$

DEFINITION
 $\mathbb{Z}_{\mathbb{F}}$ by extension
 3.4

Within field $\mathbb{F}$, let the following:

1. $0_{\mathbb{F}}$: Additive identity
2. $-x$: Additive inverse

We define the set:

$$\mathbb{Z}_{\mathbb{F}} = \{0_{\mathbb{F}}\} \cup \mathbb{N}_{\mathbb{F}} \cup \{-x_{\mathbb{F}}, \forall n_{\mathbb{F}} \in \mathbb{N}_{\mathbb{F}}\}$$

DEFINITION
 $\mathbb{Q}_{\mathbb{F}}$ by operation
 3.5

Within field $\mathbb{F}$, let the following:

1. $a \in \mathbb{F}$
2. b^{-1} (multiplicative inverse of b) $\in \mathbb{F}$

we define the set:

$$\mathbb{Q}_{\mathbb{F}} := \left\{ \frac{a}{b} : a \in \mathbb{Z}_{\mathbb{F}}, b \in \mathbb{N}_{\mathbb{F}} \right\}$$

Starting similarly to Theorem 2.7 (Archimedean property) we use the inductive approach to reach $\mathbb{N}_{\mathbb{F}}$. Fancy but not complicated at all. Later, we just extend integers in $\mathbb{F}$ to build by $\mathbb{Z}_{\mathbb{F}}$ to conclude defining $\mathbb{Q}_{\mathbb{F}}$ with the result of operating arbitrary elements $a, b \in \mathbb{F}$ as $a.b^{-1}$.

Notice how we only used existent elements from $\mathbb{F}$ computed using the field's operations and nothing else. It is important to mention that you as the reader don't know yet the usage of these new sets, although we could make a good guess.

Now comes the tricky part for us. One could think that the next steps involve 1. to prove that the irrational (in $\mathbb{F}$) is in fact buildable from $\mathbb{Q}_{\mathbb{F}}$, and 2. to prove that there is an isomorphism between this result and the $\mathbb{R}$ we know.

That won't be the case. If we would go down that path, we would just be creating an arbitrary function Φ that maps a generic $\mathbb{F}$ with $\mathbb{R}$ perfectly, losing all progress we made constructing these specific sets one by one (for a reason). The real strategy here is to first prove that Φ exists and *then* to try to extend this function to make it follow all expected properties in the irrationals. Buckle up.

3.3 Isomorphism in $\mathbb{Q}_S$

DEFINITION	Let $\phi : \mathbb{Q} \rightarrow \mathbb{Q}_{\mathbb{F}}$ defined as:	
ϕ function		$\phi\left(\frac{m}{n}\right) := \frac{m_{\mathbb{F}}}{n_{\mathbb{F}}}$
3.6	Where:	
	$m, n \in \mathbb{Q} \ (m \in \mathbb{Z}, n \in \mathbb{Z}^+)$ $m_{\mathbb{F}}, n_{\mathbb{F}} \in \mathbb{Q}_{\mathbb{F}}$	

LEMMA	$\phi : \mathbb{Q} \rightarrow \mathbb{Q}_{\mathbb{F}}$	
3.3		$\phi\left(\frac{m}{n}\right) := \frac{m_{\mathbb{F}}}{n_{\mathbb{F}}}$
	is an homomorphism	

PROOF For $m_1/n_1, m_2/n_2 \in \mathbb{Q}$

$$1. \phi(x + y) = \phi(x) + \phi(y)$$

$$\begin{aligned} \phi(m_1/n_1 + m_2/n_2) &= \phi\left(\frac{m_1.n_2 + m_2.n_1}{n_1.n_2}\right) \\ &= \frac{(m_1.n_2 + m_2.n_1)_{\mathbb{F}}}{(n_1.n_2)_{\mathbb{F}}} \\ &= \frac{(m_1.n_2)_{\mathbb{F}} + (m_2.n_1)_{\mathbb{F}}}{(n_1)_{\mathbb{F}}.(n_2)_{\mathbb{F}}} \\ &= \frac{(m_1)_{\mathbb{F}}.(n_2)_{\mathbb{F}} + (m_2)_{\mathbb{F}}.(n_1)_{\mathbb{F}}}{(n_1)_{\mathbb{F}}.(n_2)_{\mathbb{F}}} \\ &= \frac{(m_1)_{\mathbb{F}}}{(n_1)_{\mathbb{F}}} + \frac{(m_2)_{\mathbb{F}}}{(n_2)_{\mathbb{F}}} \end{aligned}$$

$$\phi(m_1/n_1 + m_2/n_2) = \phi(m_1/n_1) + \phi(m_2/n_2)$$

$$2. \phi(x.y) = \phi(x).\phi(y)$$

$$\begin{aligned} \phi(m_1/n_1.m_2/n_2) &= \phi\left(\frac{m_1.m_2}{n_1.n_2}\right) \\ &= \frac{(m_1.m_2)_{\mathbb{F}}}{(n_1.n_2)_{\mathbb{F}}} \\ &= \frac{(m_1)_{\mathbb{F}}.(m_2)_{\mathbb{F}}}{(n_1)_{\mathbb{F}}.(n_2)_{\mathbb{F}}} \\ &= \frac{(m_1)_{\mathbb{F}}}{(n_1)_{\mathbb{F}}} . \frac{(m_2)_{\mathbb{F}}}{(n_2)_{\mathbb{F}}} \end{aligned}$$

$$\phi(m_1/n_1.m_2/n_2) = \phi(m_1/n_1).\phi(m_2/n_2)$$

Finally, ϕ is an homomorphism. □

Now comes the monster. In order to get the isomorphism, we need to make sure ϕ preserves order to prove injectivity by Lemma 3.1, and this step alone takes several previous lemmas to cover it fully, and of course we will see them all. Just remember we are having fun.

LEMMA
Ordering in $\mathbb{N}_{\mathbb{F}}$
3.4

For $n, k \in \mathbb{N}, n_{\mathbb{F}}, k_{\mathbb{F}} \in \mathbb{N}_{\mathbb{F}}$

1. If $n = k \rightarrow n_{\mathbb{F}} = k_{\mathbb{F}}$
2. If $n < k \rightarrow n_{\mathbb{F}} < k_{\mathbb{F}}$
3. If $n > k \rightarrow n_{\mathbb{F}} > k_{\mathbb{F}}$

PROOF

1. $n = k \rightarrow n_{\mathbb{F}} = k_{\mathbb{F}}$

$$n_{\mathbb{F}} := \sum_1^n 1_F = \sum_1^k 1_F = n_{\mathbb{F}}$$

2. $n < k \rightarrow n_{\mathbb{F}} < k_{\mathbb{F}}$

$$n < k \Leftrightarrow \exists p \in \mathbb{N} : n + p = k$$

Now, $k_{\mathbb{F}}$ can be expressed as:

$$k_{\mathbb{F}} := \sum_1^k 1_F = \sum_1^n 1_F + \sum_1^p 1_F = n_{\mathbb{F}} + p_{\mathbb{F}}$$

Since $p > 1$, using order preservation in addition for fields (Definition 2.5) we have:

$$k_{\mathbb{F}} = n_{\mathbb{F}} + p_{\mathbb{F}} > n_{\mathbb{F}} + 0_{\mathbb{F}} > n_{\mathbb{F}}$$

3. $n > k \rightarrow n_{\mathbb{F}} > k_{\mathbb{F}}$

$$n > k \Leftrightarrow k < n$$

From 2.

$$n_{\mathbb{N}} > k_{\mathbb{N}}$$

□

LEMMA
Ordering in $\mathbb{Z}_{\mathbb{F}}$
3.5

For $n, k \in \mathbb{Z}, n_{\mathbb{F}}, k_{\mathbb{F}} \in \mathbb{Z}_{\mathbb{F}}$

1. If $n < k \rightarrow n_{\mathbb{F}} < k_{\mathbb{F}}$
2. If $n = k \rightarrow n_{\mathbb{F}} = k_{\mathbb{F}}$
3. If $n > k \rightarrow n_{\mathbb{F}} > k_{\mathbb{F}}$

PROOF

1. $n = k \rightarrow n_{\mathbb{F}} = k_{\mathbb{F}}$

(a) $n, k \geq 0$: Same as the proof for $\mathbb{N}_{\mathbb{F}}$

(b) $n, k < 0$:

$$n = k \leftrightarrow -k = -n \rightarrow -k, -n \in \mathbb{N}$$

We now proceed exactly as the proof for $\mathbb{N}_{\mathbb{F}}$

2. $n < k \rightarrow n_{\mathbb{F}} < k_{\mathbb{F}}$, similarly to $\mathbb{N}_{\mathbb{F}}$:

$$n < k \leftrightarrow \exists p \in \mathbb{N} : n + p = k$$

(a) $n \geq 0$: Since $n \in \mathbb{N} \rightarrow (n + p)_{\mathbb{F}} = n_{\mathbb{F}} + p_{\mathbb{F}}$

(b) $n < 0$: Labeling $n = -t$ just to indicate it is negative by sign:

$$\rightarrow n = -t \leftrightarrow n_{\mathbb{F}} = -t_{\mathbb{F}} \wedge (-t) + p = k$$

$$\rightarrow p = k + t$$

$$\rightarrow p_{\mathbb{F}} = k_{\mathbb{F}} + t_{\mathbb{F}}$$

$$\rightarrow p_{\mathbb{F}} = k_{\mathbb{F}} - n_{\mathbb{F}}$$

$$\rightarrow n_{\mathbb{F}} + p_{\mathbb{F}} = k_{\mathbb{F}}$$

$$\rightarrow n_{\mathbb{F}} + p_{\mathbb{F}} = k_{\mathbb{F}} = (n + p)_{\mathbb{F}}$$

Therefore, we proved for any case of $n, k \in \mathbb{Z} : n < k \rightarrow n_{\mathbb{F}} < k_{\mathbb{F}}$

3. $n > k \rightarrow n_{\mathbb{F}} > k_{\mathbb{F}}$

Labeling accordingly: $n > k \leftrightarrow -n' > -k' \leftrightarrow n' < k' (n', k' \in \mathbb{N})$

With this we just proceed as in 2

□

So far so “good”. Now, before moving to $\mathbb{Q}$ we need an extra tool for operating multiplications in $\mathbb{Z}_{\mathbb{F}}$.

LEMMA
Multiplication in $\mathbb{Z}_{\mathbb{F}}$
3.6

For any $n, k \in \mathbb{Z}$

$$(n.k)_{\mathbb{F}} = n_{\mathbb{F}}.k_{\mathbb{F}} \in \mathbb{F}$$

PROOF

1. $n, k > 0$:

We have:

$$(n.k)_{\mathbb{F}} = \sum_1^{n.k} 1_{\mathbb{F}} = \sum_1^n \sum_1^k 1_{\mathbb{F}} = \sum_1^n k_{\mathbb{F}} = n_{\mathbb{F}}.k_{\mathbb{F}}$$

Which is just adding the additive neutral $n.k$ times.

2. $n > 0, k < 0$:

Writing k as $-m$ to denote negativeness, we have:

$$(n.k)_{\mathbb{F}} = (n(-m))_{\mathbb{F}} = (-n.m)_{\mathbb{F}} = -(n.m)_{\mathbb{F}}$$

From 1:

$$(n.k)_{\mathbb{F}} = -n_{\mathbb{F}}.m_{\mathbb{F}} = n_{\mathbb{F}}. -m_{\mathbb{F}} = n_{\mathbb{F}}.k_{\mathbb{F}}$$

□

LEMMA
Ordering in $\mathbb{Q}$
3.7

Let $\frac{m_1}{n_1}, \frac{m_2}{n_2} \in \mathbb{Q}$

$$\frac{m_1}{n_1} < \frac{m_2}{n_2} \rightarrow \left(\frac{m_1}{n_1}\right)_{\mathbb{F}} < \left(\frac{m_2}{n_2}\right)_{\mathbb{F}}$$

PROOF Arbitrary locking $n_1, n_2 \in \mathbb{N}$ without losing generalization, we have:

$$\frac{m_1}{n_1} < \frac{m_2}{n_2} \leftrightarrow m_1 \cdot n_2 < m_2 \cdot n_1$$

Maintaining the inequality direction Now, from Lemma 3.5:

$$m_1 \cdot n_2 < m_2 \cdot n_1 \leftrightarrow (m_1 \cdot n_2)_{\mathbb{F}} < (m_2 \cdot n_1)_{\mathbb{F}}$$

Next, from Lemma 3.6:

$$(m_1 \cdot n_2)_{\mathbb{F}} < (m_2 \cdot n_1)_{\mathbb{F}} \leftrightarrow (m_1)_{\mathbb{F}} \cdot (n_2)_{\mathbb{F}} < (m_2)_{\mathbb{F}} \cdot (n_1)_{\mathbb{F}}$$

Now, since we fixed $n_1, n_2 \in \mathbb{N}$, we can use $(n_1 \cdot n_2)_{\mathbb{F}}$ to divide the expression:

$$(m_1)_{\mathbb{F}} \cdot (n_2)_{\mathbb{F}} < (m_2)_{\mathbb{F}} \cdot (n_1)_{\mathbb{F}} \leftrightarrow \frac{(m_1)_{\mathbb{F}} \cdot (n_2)_{\mathbb{F}}}{(n_1 \cdot n_2)_{\mathbb{F}}} \cdot \frac{(m_2)_{\mathbb{F}} \cdot (n_1)_{\mathbb{F}}}{(n_1 \cdot n_2)_{\mathbb{F}}}$$

Next, we just distribute and simplify:

$$\frac{(m_1)_{\mathbb{F}} \cdot (n_2)_{\mathbb{F}}}{(n_1 \cdot n_2)_{\mathbb{F}}} \cdot \frac{(m_2)_{\mathbb{F}} \cdot (n_1)_{\mathbb{F}}}{(n_1 \cdot n_2)_{\mathbb{F}}} \leftrightarrow \frac{(m_1)_{\mathbb{F}}}{(n_1)_{\mathbb{F}}} \cdot \frac{(m_2)_{\mathbb{F}}}{(n_1)_{\mathbb{F}}}$$

Finally, we have:

$$\frac{m_1}{n_1} < \frac{m_2}{n_2} \leftrightarrow \frac{(m_1)_{\mathbb{F}}}{(n_1)_{\mathbb{F}}} \cdot \frac{(m_2)_{\mathbb{F}}}{(n_1)_{\mathbb{F}}}$$

$$\frac{m_1}{n_1} < \frac{m_2}{n_2} \leftrightarrow \left(\frac{m_1}{n_1}\right)_{\mathbb{F}} < \left(\frac{m_2}{n_2}\right)_{\mathbb{F}}$$

□

Now if we recall, we did all of this as a previous step for proving the injectivity of ϕ . We can make it happen now.

LEMMA
3.8

$$\phi : \mathbb{Q}_{\mathbb{F}} \rightarrow \mathbb{Q}$$

$$\phi\left(\frac{m}{n}\right) := \frac{m_{\mathbb{F}}}{n_{\mathbb{F}}}$$

is injective

PROOF Let $\frac{m_1}{n_1}, \frac{m_2}{n_2} \in \mathbb{Q}$, such that:

$$\frac{m_1}{n_1} < \frac{m_2}{n_2}$$

By Lemma 3.7, we have:

$$\frac{m_1}{n_1} < \frac{m_2}{n_2} \rightarrow \left(\frac{m_1}{n_1}\right)_{\mathbb{F}} < \left(\frac{m_2}{n_2}\right)_{\mathbb{F}}$$

By definition of ϕ , we can replace:

$$\frac{m_1}{n_1} < \frac{m_2}{n_2} \rightarrow \frac{(m_1)_{\mathbb{F}}}{(n_1)_{\mathbb{F}}} < \frac{(m_2)_{\mathbb{F}}}{(n_2)_{\mathbb{F}}}$$

$$\frac{m_1}{n_1} < \frac{m_2}{n_2} \rightarrow \phi\left(\frac{m_1}{n_1}\right) < \phi\left(\frac{m_2}{n_2}\right)$$

$\Rightarrow \phi : \mathbb{Q} \rightarrow \mathbb{Q}_{\mathbb{F}}$ preserves order.

Finally, from Lemma 3.1 (Injectivity by order preservation) we assert:

$\Rightarrow \phi : \mathbb{Q} \rightarrow \mathbb{Q}_{\mathbb{F}}$ is injective.

□

LEMMA
3.9

$\phi : \mathbb{Q}_{\mathbb{F}} \rightarrow \mathbb{Q}$

$$\phi\left(\frac{m}{n}\right) := \frac{m_{\mathbb{F}}}{n_{\mathbb{F}}}$$

is surjective

PROOF

By set construction (Lemma 3.2) we defined $\phi : \mathbb{Q}_{\mathbb{F}} \rightarrow \mathbb{Q}$, meaning that:

$$\mathbb{Q} = \text{Im}(\mathbb{Q}_{\mathbb{F}})$$

Then

ϕ is surjective.

□

This completes the construction.

LEMMA
3.10

$$\phi : \mathbb{Q}_{\mathbb{F}} \rightarrow \mathbb{Q}$$

$$\phi\left(\frac{m}{n}\right) := \frac{m_{\mathbb{F}}}{n_{\mathbb{F}}}$$

is an isomorphism

PROOF

1. By Lemma 3.3, ϕ is an homomorphism.

2. By Lemma 3.8 and Lemma 3.9, ϕ is bijective

Then, by Definition 3.2

$\Rightarrow \phi$ is an isomorphism.

□

Securing this is a big step, and recalling our strategy we now need to use this isomorphism ϕ in $\mathbb{Q}_{\mathbb{F}} \rightarrow \mathbb{Q}$, and somehow transform it into a Φ that extends this to the whole $\mathbb{F} \rightarrow \mathbb{R}$.

Although this has been a long ride, personally I don't think this was a complex construction, or a hard-to-follow implementation of ϕ . Everything just landed in place.

The tricky part begins when we introduce Φ and try to understand what it really is. Still, I wouldn't include anything that I wouldn't find clear or satisfactory from a student's perspective (keep in mind I'm also learning while writing). If something feels confusing or lacks intuition, I prefer to leave it out entirely and try again.

3.4 Isomorphism in $\mathbb{R}$

We have successfully constructed an isomorphism $\phi : \mathbb{Q}_{\mathbb{F}} \rightarrow \mathbb{Q}$ that preserves all the important conditions (order, addition and multiplication), but our main goal is to extend this to all $\mathbb{R}$, starting from all $\mathbb{F}$.

So we expect a:

$$\Phi : \mathbb{F} \rightarrow \mathbb{R}$$

Such that:

1. Matches the former ϕ for all $\mathbb{Q}_{\mathbb{F}}$ into $\mathbb{Q}$
2. Remains an isomorphism
3. Preserves the field's structure (order, addition and multiplication)

The intuition to create Φ was actually mentioned previously, on Example 2.1. We talked about how a subset S of an ordered field $\mathbb{Q}$ was bounded above by $\sqrt{2}$. The example talks more about how this S won't have a supremum, but the key idea we will be using is how this particular S was **bounded** by an irrational like $\sqrt{2}$. This is the relationship we need to apply to link irrational and rational numbers: **the first one bounds the second one**

With this idea, we can intuit that any **real number is actually defined by the numbers below it**. Take $\sqrt{2}$ for example:

1. $\sqrt{2}$ is greater than all the rationals q where $q < \sqrt{2}$: 1, 1.4, 1.41, 1.414, ...

2. $\sqrt{2}$ is less than all the rationals q where $q > \sqrt{2}$: 2, 1.5, 1.42, 1.4145, ...

The fact is that $\sqrt{2}$ is the only number placed in the boundary between these two sequences.

Formally:

$$\sqrt{2} := \sup\{q \in \mathbb{Q} : q < \sqrt{2}\}$$

Relating an irrational number, with all the rationals below it.

DEFINITION

Φ function

3.7

For any $\alpha \in \mathbb{R}$, define $\Phi : \mathbb{R} \rightarrow \mathbb{F}$ as

$$\Phi(\alpha) := \sup\{\phi(q) : q \in \mathbb{Q}, q < \alpha\}$$

This new function Φ using the former isomorphism ϕ defined in $\mathbb{Q}$ does the exact same thing that $\sqrt{2}$ did in our example: bounds $\mathbb{Q}$. Since our main focus is to somehow extend the properties of the original ϕ , let us take the time to emphasize why this new Φ is not just an arbitrary function.

LEMMA

3.11

For $r \in \mathbb{Q}$

$$\Phi(r) = \phi(r)$$

PROOF

First, let us check the definition:

$$\Phi(\alpha) := \sup\{\phi(q) : q \in \mathbb{Q}, q < \alpha\}$$

If α happens to be a rational number r , it means that the supremum of all the rational values before this particular r are limited by $\phi(r)$, and this is the *Least Upper Bound* of the set.

1. $\phi(r)$ bounds $\{\phi(q) : q \in \mathbb{Q}, q < r\}$

Knowing that ϕ preserves order:

$$\forall q, q < r \leftrightarrow \phi(q) < \phi(r)$$

$$\leftrightarrow \phi(q) \leq \phi(r)$$

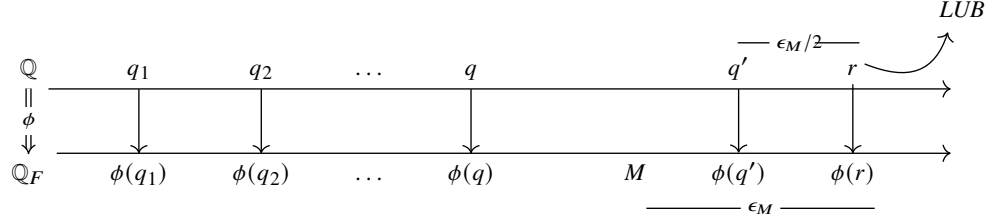
$\Rightarrow \phi(r)$ bounds the set.

□

PROOF

2. $\phi(r)$ is the L.U.B. of $\{\phi(q) : q \in \mathbb{Q}, q < r\}$

let us visualize the approach



Assume M is an arbitrary upper bound, trying to make $\phi(r)$ not the *L.U.B.*

$$\rightarrow M < \phi(r) \rightarrow \phi(r) - M > 0$$

Let us denote ϵ_M as the distance between this assumed M and $\phi(r)$.

Now, let us not forget that by density in $\mathbb{Q}$ we are able to locate an arbitrary rational number between two other arbitrary rationals, or even better for our use case: we can position an arbitrary rational q' as close as we want to any other arbitrary rational.

Developing that idea:

$$\forall r \in \mathbb{Q}, \exists q' \in \mathbb{Q} : q' - \epsilon \leq r, \epsilon \in \mathbb{Q}$$

Now, $M \in \mathbb{Q}_F$, but the density property we are using for q' is in $\mathbb{Q}$. Luckily, we already proved that ϕ is ordered:

$$\begin{aligned} \forall r \in \mathbb{Q}, \exists q' \in \mathbb{Q} / \forall \epsilon > 0 : q' - \epsilon \leq r \\ \rightarrow \phi(q') - \phi(\epsilon) \leq \phi(r) \end{aligned}$$

Now that we are entirely in $\mathbb{Q}_F$ we will conveniently take this distance $\phi(\epsilon)$ as $\epsilon_M/2$, just to note that this will be smaller than the distance between M and r . No matter how close we assume M is to $\phi(r)$, by density in $\mathbb{Q}$ we can make sure that we can take half that distance (or a quarter, or a tenth, doesn't matter) to locate another image closer to $\phi(r)$.

$$\rightarrow M < \phi(q') < \phi(r) \rightarrow M \neq \text{Upper bound}$$

$$\Rightarrow \Phi(r) := \sup\{\phi(q) : q \in \mathbb{Q}, q < r\} = \phi(r), \forall r \in \mathbb{Q}$$

□

Taking the time to prove this mapping for rationals wasn't a waste of time after all, since it confirms that our intuition about the definition of Φ isn't arbitrary, but respects the structure we already built.

So far, our definition for Φ :

$$\Phi(\alpha) := \sup\{\phi(q) : q \in \mathbb{Q}, q < \alpha\}$$

Uses an internal set $S(\alpha) = \{\phi(q) : q \in \mathbb{Q}, q < \alpha\}$ that we can use to simplify our readability a bit:

$$\Phi(\alpha) := \sup(S(\alpha))$$

Now, to make Φ to make sense we need to make sure that S makes sense, and if we want to obtain the supremum of a poorly-defined set, the whole definition crumbles.

Next, we will review how $S(\alpha)$ is well defined to avoid any issues further down.

LEMMA	Set S :	$\Phi(\alpha) := \sup\{S(\alpha)\} = \sup\{\phi(q) : q \in \mathbb{Q}, q < \alpha\}$
3.12	is well defined in $\mathbb{R}$	

PROOF To prove this, what matters the most is that S is well-defined in $\mathbb{F}$. That automatically guarantees that Φ makes sense in $\mathbb{R}$.

1. $S(\alpha)$ is not empty:

Using Theorem 2.7 (Archimedean theorem):

$$\begin{aligned}\forall \alpha \in \mathbb{Q}, \exists n \in \mathbb{N} : n > \alpha \\ \rightarrow n > |\alpha| \\ \rightarrow -n < \alpha, -n \in \mathbb{Q}\end{aligned}$$

Identifying $q = -n$ in S , we have a rational that is bounded by α , meaning that $\phi(-n) \in S(\alpha)$

2. $S(\alpha)$ is bounded above:

Again, using Theorem 2.7:

$$\forall \alpha \in \mathbb{Q}, \exists n \in \mathbb{N} : n > \alpha$$

Then, for any rational $q < \alpha$

$$\begin{aligned}n > \alpha > q \\ \rightarrow q < n \\ \rightarrow \phi(q) < \phi(n) = n_{\mathbb{F}}\end{aligned}$$

So $n_{\mathbb{F}}$ is an upper bound of S , for any $q < \alpha$

$\rightarrow S(\alpha)$ is well-defined in $\mathbb{F}$

$\rightarrow \Phi(\alpha) := \sup\{S(\alpha)\}$ is well-defined in $\mathbb{R}$

□

Now, while studying this a question came to me: if we already know that $\phi : \mathbb{Q} \rightarrow \mathbb{Q}_{\mathbb{F}}$ exists, why do we need to prove $S(\alpha)$ is non-empty? Can't we just pick any rational and map it?

The issue is that the set $S(\alpha)$ is not just any images of rationals, it's specifically: $S(\alpha) = \{\phi(q) : q \in \mathbb{Q} \text{ AND } q < \alpha\}$. The constraint $q < \alpha$ obliges us to prove that *there exists at least one rational q with $q < \alpha$* .

LEMMA	
Order-Preservation	For $\alpha, \beta \in \mathbb{R}$
3.13	$\alpha < \beta \rightarrow \Phi(\alpha) < \Phi(\beta)$

PROOF let us start with two evaluations of set S :

- $S(\alpha) := \{\phi(q) : q \in \mathbb{Q}, q < \alpha\}$
- $S(\beta) := \{\phi(q) : q \in \mathbb{Q}, q < \beta\}$

If $\alpha < \beta$, then for any $\phi(q') \in S(\alpha)$:

$$\begin{aligned} \phi(q) \in S(\alpha) &\leftrightarrow \phi(q') \in \{\phi(q) : q \in \mathbb{Q}, q < \alpha\} \\ &\leftrightarrow \phi(q') \in \{\phi(q) : q \in \mathbb{Q}, q < \alpha < \beta\} \\ &\leftrightarrow \phi(q') \in \{\phi(q) : q \in \mathbb{Q}, q < \beta\} \\ &\leftrightarrow \phi(q') \in S(\beta) \end{aligned}$$

Then, going back to the statement:

$$\begin{aligned} \alpha < \beta &\Rightarrow S(\alpha) \subseteq S(\beta) \\ &\Rightarrow \sup(S(\alpha)) \leq \sup(S(\beta)) \\ &\Rightarrow \Phi(\alpha) \leq \Phi(\beta) \end{aligned}$$

Now, to prove that the images of α and β in Φ are strictly less than and not equal, we use the density of $\mathbb{Q}$ in $\mathbb{R}$ to locate arbitrary rationals r_α, r_β .

$$\begin{aligned} \forall \alpha, \beta \in \mathbb{R}, \exists r_\alpha, r_\beta \in \mathbb{Q} : \\ \alpha < \beta \rightarrow \alpha < r_\alpha < r_\beta < \beta \end{aligned}$$

Since $r_\beta < \beta \rightarrow \phi(r_\beta) \in S(\beta)$ by the definition of S . Now, knowing that for any element of $S(\beta)$ it will be less or equal to its supremum, we have:

$$\begin{aligned} \phi(r_\beta) &\leq \sup(S(\beta)) \\ \phi(r_\beta) &\leq \Phi(\beta) \end{aligned}$$

Now in the middle, by order preservation:

$$r_\alpha < r_\beta \leftrightarrow \phi(r_\alpha) < \phi(r_\beta)$$

And finally, since $\alpha < r_\alpha \rightarrow S(\alpha) \subseteq S(r_\alpha)$. Then:

$$\begin{aligned} S(\alpha) &\subseteq S(r_\alpha) \\ &\rightarrow \Phi(\alpha) \leq \Phi(r_\alpha) \\ &\rightarrow \Phi(\alpha) \leq \phi(r_\alpha) \text{ (Since } r_\alpha \in \mathbb{Q}) \end{aligned}$$

Joining all results:

$$\begin{aligned} \Phi(\alpha) &\leq \phi(r_\alpha) < \phi(r_\beta) \leq \Phi(\beta) \\ &\Rightarrow \Phi(\alpha) < \Phi(\beta) \end{aligned}$$

Finally, $\alpha < \beta \Rightarrow \Phi(\alpha) < \Phi(\beta)$. □

Now comes the final sprint to prove that Φ maps $\mathbb{F}$ and $\mathbb{R}$ perfectly. Just as we did with ϕ , we need to prove isomorphism by proving bijection and homomorphism. From this last part, the only tricky step is for surjectiveness, since it uses a method that we haven't seeing before. We are almost there!

LEMMA
Injectivity in Φ
3.14

Given $\Phi : \mathbb{R} \rightarrow \mathbb{F}$, for $\alpha, \beta \in \mathbb{R}$:

$$\alpha \neq \beta \rightarrow \Phi(\alpha) \neq \Phi(\beta)$$

PROOF

From proven order preservation in Lemma 3.13, if $\alpha \neq \beta$

1. $\alpha < \beta \rightarrow \Phi(\alpha) < \Phi(\beta) \rightarrow \Phi(\alpha) \neq \Phi(\beta)$
2. $\alpha > \beta \rightarrow \Phi(\alpha) > \Phi(\beta) \rightarrow \Phi(\alpha) \neq \Phi(\beta)$

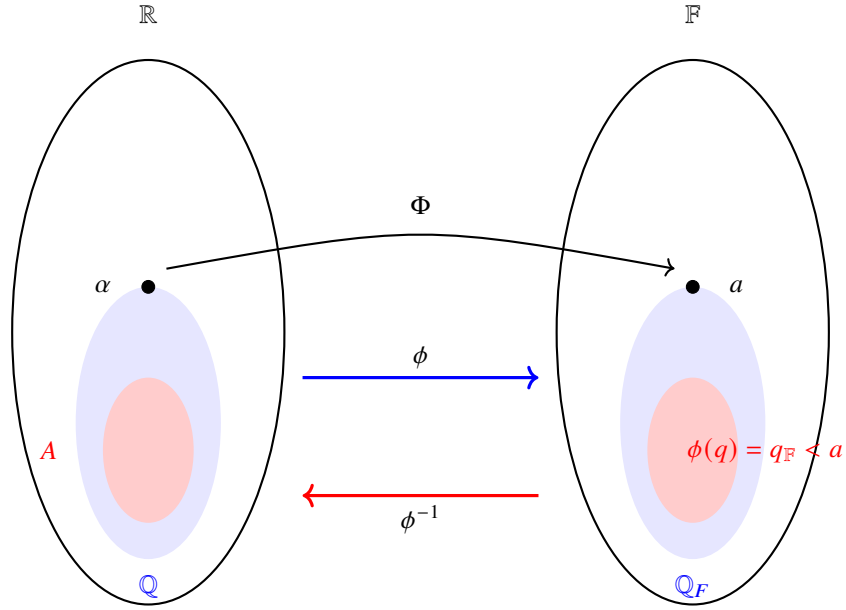
□

LEMMA
Surjectivity in Φ
3.15

Given $\Phi : \mathbb{R} \rightarrow \mathbb{F}$:

$$\forall a \in \mathbb{F}, \exists \alpha \in \mathbb{R} : \Phi(\alpha) = a$$

PROOF Let us support the following ideas on the next diagram:



Picking an arbitrary $a \in \mathbb{F}$, we consider the set of rational elements of $\mathbb{F}$ that are strictly smaller than a :

$$\{q_{\mathbb{F}} \in \mathbb{Q}_{\mathbb{F}} : q_{\mathbb{F}} < a\}.$$

Now, since $\phi : \mathbb{Q} \rightarrow \mathbb{Q}_{\mathbb{F}}$ is an isomorphism (and hence injective), its inverse ϕ^{-1} exists. Applying ϕ^{-1} to the set above, we define

$$\begin{aligned} A &= \phi^{-1}(\{q_{\mathbb{F}} \in \mathbb{Q}_{\mathbb{F}} : q_{\mathbb{F}} < a\}) \\ &= \phi^{-1}(\{\phi(q) : q \in \mathbb{Q}, \phi(q) < a\}) \\ &= \{q \in \mathbb{Q} : \phi(q) < a\}. \end{aligned}$$

Now let us consider $\alpha = \sup(A)$. We first show that α exists.

1. A is non-empty:

Since $\mathbb{Q}_{\mathbb{F}}$ is dense in $\mathbb{F}$ and $a \in \mathbb{F}$, there exists $q_{\mathbb{F}} \in \mathbb{Q}_{\mathbb{F}}$ such that $q_{\mathbb{F}} < a$. By surjectivity of ϕ onto $\mathbb{Q}_{\mathbb{F}}$, there exists $q \in \mathbb{Q}$ with $\phi(q) = q_{\mathbb{F}}$. Hence $\phi(q) < a$ and therefore $q \in A$.

2. A is bounded above:

Using the Archimedean property in $\mathbb{F}$, there exists $n_{\mathbb{F}} \in \mathbb{N}_{\mathbb{F}}$ such that $a < n_{\mathbb{F}}$. Since ϕ is order-preserving and maps $\mathbb{N}$ onto $\mathbb{N}_{\mathbb{F}}$, there exists $n \in \mathbb{N}$ with $\phi(n) = n_{\mathbb{F}}$.

For every $q \in A$ we have $\phi(q) < a < n_{\mathbb{F}} = \phi(n)$, and by order preservation of ϕ this implies $q < n$. Hence n is an upper bound for A .

Therefore, $\alpha = \sup(A)$ exists. □

PROOF Now, what we want to achieve is that, given this arbitrary $a \in \mathbb{F}$, we reach a real number α such that $\Phi(\alpha) = a$. To do this, we show that the set A constructed above coincides with all rationals strictly smaller than its supremum.

Claim:

$$\{q \in \mathbb{Q} : q < \alpha\} = A.$$

• $\subseteq$:

Let $q \in \mathbb{Q}$ with $q < \alpha$. Since $\alpha = \sup(A)$, q cannot be an upper bound for A . Hence there exists $q' \in A$ such that

$$q < q' \leq \alpha.$$

Since ϕ is order-preserving, we have

$$\phi(q) < \phi(q').$$

But $q' \in A$ implies $\phi(q') < a$, and therefore

$$\phi(q) < a,$$

which shows that $q \in A$.

• $\supseteq$:

Let $q \in A$. By definition, $\phi(q) < a$. Since $\alpha = \sup(A)$, every element of A is smaller than or equal to α , and therefore

$$q < \alpha.$$

Thus $q \in \{q \in \mathbb{Q} : q < \alpha\}$.

Hence,

$$\{q \in \mathbb{Q} : q < \alpha\} = A.$$

So far we have shown that:

1. $\alpha = \sup(A)$ exists,
2. $A = \{q \in \mathbb{Q} : q < \alpha\}$.

Combining these, we obtain

$$\sup(A) = \sup\{q \in \mathbb{Q} : q < \alpha\}.$$

By definition of Φ , we have

$$\begin{aligned} \Phi(\alpha) &= \sup\{\phi(q) : q \in \mathbb{Q}, q < \alpha\} \\ &= \sup\{\phi(q) : q \in A\}. \end{aligned}$$

But by construction of A ,

$$\{\phi(q) : q \in A\} = \{q_{\mathbb{F}} \in \mathbb{Q}_{\mathbb{F}} : q_{\mathbb{F}} < a\}.$$

Since $\mathbb{Q}_{\mathbb{F}}$ is dense in $\mathbb{F}$ and a is an upper bound for this set, it follows that

$$\sup\{q_{\mathbb{F}} \in \mathbb{Q}_{\mathbb{F}} : q_{\mathbb{F}} < a\} = a.$$

Therefore,

$$\Phi(\alpha) = a,$$

which proves that Φ is surjective. □

Now that this is done, we can conclude that Φ is a bijection. Next we will just cover the requirements to prove homomorphism and finish the whole demonstration. But first, a handy property:

LEMMA
3.16

Let sets X, Y , then:

$$\sup(X + Y) = \sup(X) + \sup(Y)$$

PROOF

As always, to prove that something is a supremum of a set, we need to prove both the bound is what we expect, and that it is in fact the *Least Upper Bound*

1. Bound:

We have:

$$\begin{aligned} X + Y &:= \{x + y, x \in X, y \in Y\} \\ &\rightarrow \forall x \in X, x \leq \sup(X) \\ &\rightarrow \forall y \in Y, y \leq \sup(Y) \\ &\rightarrow x + y \leq \sup(X) + \sup(Y) \\ &\Rightarrow \sup(X) + \sup(Y) \text{ bounds } X + Y \end{aligned}$$

2. $\sup(X) + \sup(Y)$ is *L.U.B*

Assume a new upper bound M so that $M < \sup(X) + \sup(Y)$. Then

$$\epsilon = (\sup X + \sup Y - M) > 0$$

Now, since $x < \sup X$ and $y < \sup Y$, $\forall x \in X, y \in Y$ respectively, we can take any small amount $\epsilon/2$ that would satisfy:

$$\begin{aligned} x + \epsilon/2 &> \sup X \rightarrow x > \sup X - \epsilon/2 \\ y + \epsilon/2 &> \sup Y \rightarrow y > \sup Y - \epsilon/2 \end{aligned}$$

Adding up:

$$x + y > \sup X + \sup Y - \epsilon = M$$

The, we found an $x + y \in X + Y$ that is not bounded by M .

$\Rightarrow M$ fails to bound $X + Y$

$\Rightarrow \sup X + \sup Y$ is the *L.U.B* of $X + Y$

□

LEMMA
Additivity of Φ
3.17

$\Phi : \mathbb{R} \rightarrow \mathbb{F}$ preserves addition:

$$\Phi(\alpha + \beta) = \Phi(\alpha) + \Phi(\beta), \quad \forall \alpha, \beta \in \mathbb{R}$$

PROOF

Claim: Recalling the definition of $S(\alpha) := \{\phi(q), q \in \mathbb{Q}, q < \alpha\}$, we will proof

$$S(\alpha + \beta) = S(\alpha) + S(\beta)$$

i. $\subseteq$:

Let $\phi(q) \in S(\alpha + \beta)$

$$\rightarrow q < \alpha + \beta \text{ (by definition of } S) \rightarrow q - \alpha < \beta$$

Next, by density in $\mathbb{Q}$, there must be an arbitrary q' such that:

$$\rightarrow q - \alpha < q' < \beta$$

$$\rightarrow_1 q - \alpha < q' \leftrightarrow q - q' < \alpha$$

$$\rightarrow_2 q' < \beta$$

$$\rightarrow_1 q - q' \in S(\alpha)$$

$$\rightarrow_2 q' \in S(\beta)$$

Finally, with some clever operations in ϕ using its property for preserving addition:

$$\phi(q) = \phi(q - q' + q') = \phi(q - q') + \phi(q')$$

$$\Rightarrow \phi(q) \in S(\alpha) + S(\beta)$$

ii. $\supseteq$:

Let $\phi(q_1) + \phi(q_2) \in S(\alpha) + S(\beta)$, where $q_1 < \alpha \wedge q_2 < \beta$

$$\rightarrow q_1 + q_2 < \alpha + \beta \wedge q_1 + q_2 \in \mathbb{Q}$$

$$\rightarrow q_1 + q_2 \in S(\alpha) + S(\beta)$$

Concluding that $S(\alpha + \beta) = S(\alpha) + S(\beta)$. Applying supremum to both sides, we have:

$$\sup(S(\alpha + \beta)) = \sup(S(\alpha) + S(\beta))$$

And from Lemma 3.16:

$$\sup(S(\alpha + \beta)) = \sup(S(\alpha)) + \sup(S(\beta))$$

$$\Rightarrow \Phi(\alpha + \beta) = \Phi(\alpha) + \Phi(\beta)$$

□

LEMMA	$\Phi : \mathbb{R} \rightarrow \mathbb{F}$ preserves multiplication:
Multiplicativity of Φ	$\Phi(\alpha \cdot \beta) = \Phi(\alpha) \cdot \Phi(\beta), \quad \forall \alpha, \beta \in \mathbb{R}$
3.18	

PROOF Before splitting into cases, note a consequence of Lemma 3.17: since Φ preserves addition,

$$\Phi(\alpha) + \Phi(-\alpha) = \Phi(\alpha + (-\alpha)) = \Phi(0)$$

Taking $\alpha = 0$ gives $\Phi(0) = 0_{\mathbb{F}}$, and substituting back:

$$\Phi(-\alpha) = -\Phi(\alpha), \quad \forall \alpha \in \mathbb{R}$$

We will use this freely below.

Case 1: $\alpha = 0$ or $\beta = 0$. Then $\alpha\beta = 0$, and

$$\Phi(\alpha)\Phi(\beta) = 0_{\mathbb{F}} = \Phi(0) = \Phi(\alpha\beta)$$

Case 2: $\alpha, \beta > 0$.

i. $\Phi(\alpha)\Phi(\beta)$: Upper Bound for $S(\alpha\beta)$

Let us take any $\phi(q) \in S(\alpha\beta)$ where $q < \alpha\beta$.

If $q \leq 0$: since $\alpha, \beta > 0$, density of $\mathbb{Q}$ in $\mathbb{R}$ lets us pick positive rationals $q_1 < \alpha$ and $q_2 < \beta$. Then $q_1q_2 > 0 \geq q$, so

$$\phi(q) \leq \phi(q_1q_2) = \phi(q_1)\phi(q_2)$$

If $q > 0$: since $\beta > 0$ and $q < \alpha\beta$, dividing by β gives $\frac{q}{\beta} < \alpha$. By density of $\mathbb{Q}$ in $\mathbb{R}$, we can locate a rational squeezed in between:

$$\frac{q}{\beta} < q_1 < \alpha, \quad q_1 \in \mathbb{Q}$$

Since $q_1 > q/\beta > 0$, dividing by q_1 gives $\frac{q}{q_1} < \beta$. Applying density once more:

$$\frac{q}{q_1} < q_2 < \beta, \quad q_2 \in \mathbb{Q}$$

Rearranging $q_2 > q/q_1$ we get $q < q_1q_2$, with $q_1 < \alpha$ and $q_2 < \beta$ (and both may be taken positive, since $\alpha, \beta > 0$). By order preservation of ϕ :

$$\phi(q) < \phi(q_1q_2) = \phi(q_1)\phi(q_2)$$

In either case, we land on rationals $q_1 < \alpha$ and $q_2 < \beta$ with $\phi(q) \leq \phi(q_1)\phi(q_2)$.

Since $q_1 < \alpha$ and $q_1 \in \mathbb{Q}$:

$$\phi(q_1) \in S(\alpha)$$

Therefore:

$$\phi(q_1) \leq \sup S(\alpha) = \Phi(\alpha)$$

Similarly, since $q_2 < \beta$ and $q_2 \in \mathbb{Q}$:

$$\phi(q_2) \leq \Phi(\beta)$$

Since q_1, q_2 were taken positive, multiplying these two inequalities preserves their direction:

$$\phi(q_1) \cdot \phi(q_2) \leq \Phi(\alpha) \cdot \Phi(\beta)$$

Chaining everything together:

$$\phi(q) \leq \phi(q_1) \cdot \phi(q_2) \leq \Phi(\alpha) \cdot \Phi(\beta)$$

This holds for every element of $S(\alpha\beta)$.

$$\Rightarrow \Phi(\alpha)\Phi(\beta) \text{ is an upper bound for } S(\alpha\beta)$$

□

PROOF ii. $\Phi(\alpha)\Phi(\beta)$: *L.U.B. of $S(\alpha\beta)$*

We need to prove that for any $q \in S(\alpha\beta)$, it will be bounded by $\Phi(\alpha) \cdot \Phi(\beta)$. This is equivalent to prove that there is an $\epsilon_{\mathbb{F}} > 0$, the smallest of them all, that makes $\phi(q) + \epsilon_{\mathbb{F}} > \Phi(\alpha) \cdot \Phi(\beta)$. Let $q = q_1 \cdot q_2$, $q_1 < \alpha \wedge q_2 < \beta$. Next, we define a distance δ small enough that:

$$q_1 + \delta > \alpha \wedge q_2 + \delta > \beta$$

Knowing this, our δ must be a positive number in $\mathbb{Q}$. So let us define it as

$$\delta \in \mathbb{Q}, 0 < \delta < 1$$

We will revisit this definition of δ later. Next we operate:

$$\begin{aligned} \rightarrow \alpha < q_1 + \delta &\leftrightarrow \Phi(\alpha) < \Phi(q_1 + \delta) \\ &\leftrightarrow \Phi(\alpha) < \Phi(q_1) + \delta \\ &\leftrightarrow \Phi(\alpha) < \phi(q_1 + \delta) \quad (q_1 + \delta \in \mathbb{Q}) \end{aligned}$$

Identically, we make the same operation for β and q_2

$$\begin{aligned} \rightarrow \Phi(\alpha) &< \phi(q_1 + \delta) \\ \rightarrow \Phi(\beta) &< \phi(q_2 + \delta) \end{aligned}$$

Multiplying both inequalities we have:

$$\begin{aligned} \Phi(\alpha) \cdot \Phi(\beta) &< \phi(q_1 + \delta) \cdot \phi(q_2 + \delta) \\ \Phi(\alpha) \cdot \Phi(\beta) &< \phi((q_1 + \delta) \cdot (q_2 + \delta)) \quad (\phi: \text{isomorphism allows multiplication distribution}) \\ \Phi(\alpha) \cdot \Phi(\beta) &< \phi(q_1 \cdot q_2 + \delta(q_1 + q_2) + \delta^2) \\ \Phi(\alpha) \cdot \Phi(\beta) &< \phi(q_1 \cdot q_2) + \phi(\delta(q_1 + q_2) + \delta^2) \end{aligned}$$

let us leave the inequality there to see more deeply what we want to achieve using δ . The intuition here is to find a value for δ that allows the inequality to hold, no matter what.

From the definition:

$$\delta < 1 \rightarrow \delta^2 < \delta$$

Also:

$$q_1 + q_2 < \alpha + \beta$$

Multiplying both:

$$\begin{aligned} \delta(q_1 + q_2) &< \delta(\alpha + \beta) \\ \delta(q_1 + q_2) + \delta^2 &< \delta(\alpha + \beta) + \delta \\ \delta(q_1 + q_2) + \delta^2 &< \delta(\alpha + \beta + 1) \end{aligned}$$

Now allow me to do a “cleanup”. let us redefine δ (still between 0 and 1) as:

$$0 < \delta < \min\left\{\frac{\epsilon}{\alpha + \beta + 1}, 1\right\}, \epsilon > 0 \in \mathbb{Q}$$

But why? Just to do a clean simplification and end up with an isolated $\epsilon \in \mathbb{Q}$, independent from anything else. $\square$

PROOF | Going back to the inequality, we notice that the terms start to unfold:

$$\begin{aligned}
 \Phi(\alpha) \cdot \Phi(\beta) &< \phi(q_1 \cdot q_2) + \phi(\delta(q_1 + q_2) + \delta^2) \\
 &\rightarrow < \phi(q) + \phi(\delta(q_1 + q_2) + \delta^2) < \phi(q) + \phi(\delta(\alpha + \beta + 1)) = \phi(q) + \phi(\epsilon) \\
 \rightarrow \Phi(\alpha) \cdot \Phi(\beta) &< \phi(q) + \phi(\epsilon) \\
 \rightarrow \Phi(\alpha) \cdot \Phi(\beta) &< \phi(q) + \epsilon_{\mathbb{F}}
 \end{aligned}$$

This means that, by choosing a correct δ dependent of α and β , we get an arbitrary positive distance (margin) ϵ , which shows that $\Phi(\alpha) \cdot \Phi(\beta)$ is *L.U.B.* of $\phi(q)$.

$$2\Phi(\alpha, \beta) = \sup\{\phi(q), q \in \mathbb{Q}, q < \alpha \cdot \beta\} = \Phi(\alpha) \cdot \Phi(\beta), \quad \forall \alpha, \beta > 0$$

Case 3: $\alpha, \beta < 0$. Then $-\alpha, -\beta > 0$, so by the case just proven:

$$\Phi((- \alpha)(- \beta)) = \Phi(- \alpha)\Phi(- \beta) = (-\Phi(\alpha))(-\Phi(\beta)) = \Phi(\alpha)\Phi(\beta)$$

Since $(- \alpha)(- \beta) = \alpha\beta$, this gives $\Phi(\alpha\beta) = \Phi(\alpha)\Phi(\beta)$.

Case 4: $\alpha > 0, \beta < 0$ (the case $\alpha < 0, \beta > 0$ is symmetric):

Write $\beta = -\gamma$ with $\gamma > 0$. Then $\alpha\beta = -(\alpha\gamma)$, and by the case $\alpha, \gamma > 0$ proven above:

$$\Phi(\alpha\gamma) = \Phi(\alpha)\Phi(\gamma)$$

So:

$$\Phi(\alpha\beta) = \Phi(-(\alpha\gamma)) = -\Phi(\alpha\gamma) = -\Phi(\alpha)\Phi(\gamma) = \Phi(\alpha)(-\Phi(\gamma)) = \Phi(\alpha)\Phi(-\gamma) = \Phi(\alpha)\Phi(\beta)$$

$$\Rightarrow \Phi(\alpha\beta) = \Phi(\alpha)\Phi(\beta), \quad \forall \alpha, \beta \in \mathbb{R}$$

□

LEMMA Homomorphism in Φ 3.19	$\Phi : \mathbb{R} \rightarrow \mathbb{F}$ is an homomorphism
---	---

PROOF	1. By Lemma 3.17, Φ preserves addition. 2. By Lemma 3.18, Φ preserves multiplication. $\Rightarrow \Phi$ is a homomorphism.
--------------	---

□

Finally, we reached Rome

LEMMA Isomorphism Φ 3.20	$\Phi : \mathbb{R} \rightarrow \mathbb{F}$ is an isomorphism
--	--

PROOF	1. By Lemma 3.14 and Lemma 3.15, Φ is bijective. 2. By Lemma 3.19, Φ is an homomorphism For any complete ordered field $\mathbb{R}$, there exists an isomorphism Φ such that:
--------------	--

$$\Phi : \mathbb{R} \rightarrow \mathbb{F}$$

□

Personally, this is the first time I have been this rigorous with a proof, covering as much detail as I felt necessary to reach the result. For me, the real reward was the joy of knowing I understood all the pieces needed to build the puzzle, and I hope you found it equally rewarding.

3.5 Consequences of completeness

Being completely honest, proving $\mathbb{R}$ uniqueness drained me completely, so it is fair to just check a few smaller extensions (consequence of this proof) to see how the visited properties are used in other, more specific lemmas. They also use cool last names to name them, so we can brag about knowing the *Weierstrass-Cauchy-DiCaprio-Cumberbatch principle of boiling water in $\mathbb{R}^6$* . It is also a great excuse to give some color to the notes with some diagrams.

DEFINITION | A function $f : \mathbb{N} \rightarrow X$ is a *sequence* of elements of X . So that:

3.8

$$f(n) := x_n, x_n \in X$$

DEFINITION | Let $X_1, X_2, \dots, X_n, \dots$ be a sequence of sets. If $X_1 \supset X_2 \supset \dots X_n \supset \dots, \forall n \in \mathbb{N}$, we say the sequence is *nested*.

3.9

LEMMA
Cauchy-Cantor
principle
3.21

Nested Interval Lemma: Let a sequence of closed intervals:

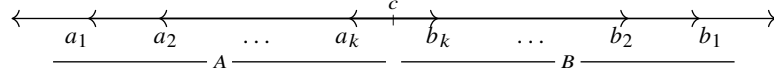
$$I_1 \supset I_2 \supset I_3 \supset \dots I_n \supset$$

Then, there exists a point $c \in \mathbb{R}$ belonging to all the intervals in the sequence. Additionally, if for any $\epsilon > 0$, there exists an interval I_k such that $\|I_k\| < \epsilon$, then the element c is unique.

PROOF Defining clearly $I_k = [a_k, b_k]$, as any arbitrary closed interval in the nested sequence. Then the sets:

$$A := \{a_k, k \in \mathbb{N}\}, B := \{b_k, k \in \mathbb{N}\}$$

Are formed using the left and right limits of the intervals mentioned.



Since A is a subset of $\mathbb{R}$, by density of $\mathbb{Q}$:

$$\exists c = \sup(A) \in \mathbb{R}, \forall a_k : a_k < c$$

Additionally (and supported by the diagram) all elements from B bound A :

$$\forall b_k \in B, \forall a_k \in A : a_k < b_k$$

Now, $c = \sup(A)$ is the least upper bound, so any element from B must be greater than c .

$$\rightarrow a_k < c < b_k$$

Finally, take special care of how *for all* a_k, b_k we found that there exists a real number c that satisfies this inequality, meaning that c belongs to any interval I_k in the nested sequence.

For the second part, condition says that for any distance $\epsilon > 0$, there is an interval I_k smaller in length than ϵ . Now, assuming there is $c_1, c_2 \in \mathbb{R}$, and for all a_k, b_k limits of the nested intervals:

$$a_k < c_1 < c_2 < b_k$$

$$\rightarrow -c_1 < -a_k \wedge c_2 < b_k$$

$$\rightarrow c_2 - c_1 < b_k - a_k$$

Then, making $b_k = a_k + \epsilon$:

$$\rightarrow c_2 - c_1 < \epsilon, \forall \epsilon > 0$$

Then, the distance between c_1 and c_2 must be less than any positive number.

$$\rightarrow c_2 - c_1 = 0$$

$$\Rightarrow c = c_2 = c_1 \text{ is unique.}$$

□

DEFINITION
Covering systems
3.10

A system $\mathcal{S} = \{X\}$ of sets X is said to cover a set Y if

$$Y \subset \bigcup_{X \in \mathcal{S}} X$$

Additionally, a subset of $\mathcal{S} = \{X\}$ will be called a *subsystem*, which is itself a system of sets of the same type.

LEMMA
Borel-Lebesgue
principle
3.22

Every system of open intervals covering a closed interval contains a finite subsystem that covers the closed interval.

PROOF What the lemma implies is just that if we consider a system $\mathcal{S} = \{\theta\}$ of open intervals (θ) that covers a closed interval $[a, b]$, there will exist a finite subsystem that also covers $[a, b]$:

$$\mathcal{S} = \{\theta\} \text{ covers } [a, b] \rightarrow \exists \mathcal{S}_S : \text{finite} / \mathcal{S}_S \text{ covers } [a, b]$$

Proving by contradiction, let's assume there is no finite subsystem that covers $[a, b] = I_1$. Since $[a, b]$ is not contained, at least one half of it (any half, doesn't matter) is not contained within $\mathcal{S}$ (if both halves were contained, we would have used a finite amount of intervals already, not following the assumption). Then, the half we choose (I_2) once again cannot be contained in a finite set of intervals from $\mathcal{S}$, otherwise we're back to the same contradiction. Notice that we can repeat this process indefinitely, ending up with the sequence:

$$I_1 \supset I_2 \supset I_3 \supset \dots \supset I_n \supset \dots$$

of nested closed intervals, where **each I_n still cannot be covered by any finite subsystem of $\mathcal{S}$** . Moreover, since each step halves the length of the previous interval, we have $|I_n| = |I_1|/2^n$, so the lengths shrink to 0 as $n \rightarrow \infty$. Let's keep this thought for a minute.

Now, we can apply Cauchy-Cantor (Lemma 3.21) into this sequence of closed intervals to assert that:

$$\begin{aligned} \exists c \in I_k, \forall I_k \text{ in the sequence} \\ \rightarrow c \in I_1 = [a, b] \end{aligned}$$

And from the definition of covering systems:

$$[a, b] \subset \bigcup_{\theta \in \mathcal{S}} \theta$$

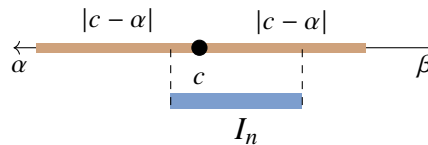
Meaning that any point from $[a, b]$ lands on a θ . Since c is a point, we get:

$$\begin{aligned} \exists \theta_i / c \in \theta_i =]\alpha, \beta[\\ \rightarrow \exists \theta_i =]\alpha, \beta[/ \alpha < c < \beta \end{aligned}$$

Now comes the trick. Since I_n keeps shrinking alongside the sequence, its length is arbitrarily smaller than any $\epsilon > 0$. We can conveniently take $\epsilon = \min\{c - \alpha, \beta - c\}$, having:

- $c \in I_n$ (Since it is in any interval I_k)
- $|I_n| < \min\{c - \alpha, \beta - c\}$

Since I_n is tied to keep c inside it, and we fixed its length to be the minimum room c still has inside our open interval $] \alpha, \beta [$, we assert:



(Picture sliding the blue interval sideways, keeping c in it. Since we chose a proper ϵ , it will fit between α and β)

$$\begin{aligned} I_n \subset]\alpha, \beta[\\ \Rightarrow \exists \theta \in \mathcal{S}, I_n \subset I_1 / I_n \subset \theta \end{aligned}$$

I_n was originally one of the many halves that were not contained within system $\mathcal{S}$, meaning that this result contradicts the initial assumption. $\square$

DEFINITION
3.11

The neighborhood of a point $x \in \mathbb{R}$ is a open interval containing the point, and the δ -neighborhood about. That is:

$$U_\delta(x) =]x - \delta, x + \delta[$$

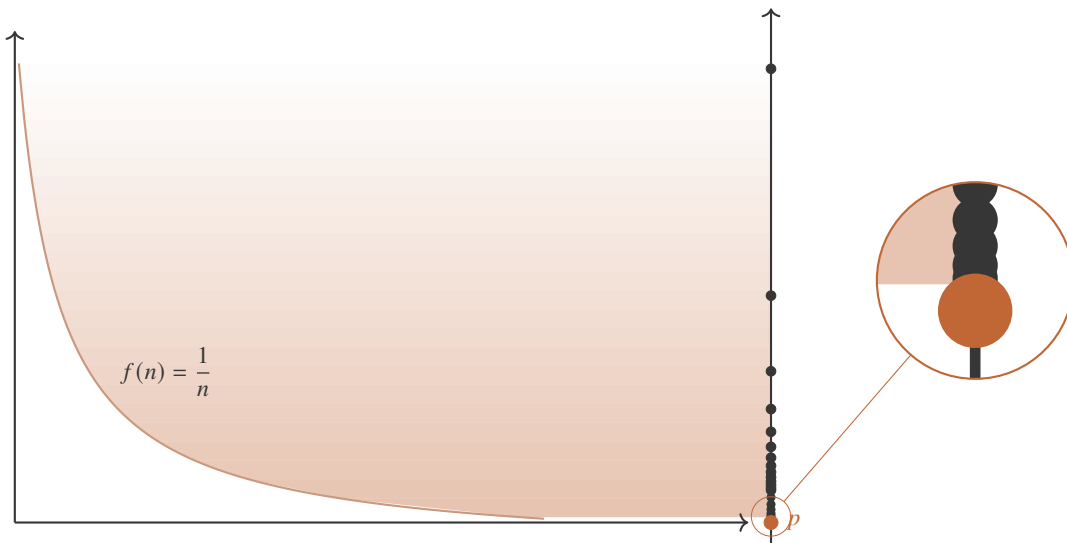
DEFINITION
Limit point
3.12

A point $p \in \mathbb{R}$ is a *limit point* of the set $X \subset \mathbb{R}$ if every neighborhood of p contains infinitely many points of X . That is:

$$p \text{ is a limit point of } X \iff \forall \delta > 0, |U_\delta(p) \cap X| = \infty$$

Think it this way: a point will be limit point of a set X if we find out that no matter where you place it (it doesn't even need to be inside the set itself), you can *check* that around that point there are elements of X . If you *check* a neighborhood big enough, of course you expect to get elements of X as you hug more and more space in $\mathbb{R}$. The important detail happens when you hug less and less space. If you zoom in and check a tiny neighborhood around the point, and *still* find elements of X there, it means that it is a limit point.

The bibliography uses the set $X = \{1/n, n \in \mathbb{N}\}$ to exemplify this, but it doesn't show the intuition behind it. Graphically, we can say that these items try to reach zero as n grows, to eventually get infinitely close to, but never reaching it. We say that 0 is the limit point of this set because no matter how tiny I force the neighborhood around it, I will always get a point from X .



Adjacent to this, we can also declare that:

- For every open interval $]a, b[$, every point of the closed interval $[a, b]$ is a limit point of it, and there are no others.
- Any point of $\mathbb{R}$ is a limit point of $\mathbb{Q}$, as we know that every open interval of the real numbers contains rational numbers.

LEMMA
Bolzano-
Weierstrass
3.23

Every bounded infinite set of real numbers has at least one limit point.

PROOF

For this, it is equivalent to prove:

$$\forall X \subset \mathbb{R}, X : \text{infinite and bounded} \Rightarrow \exists p \in \mathbb{R} : \text{limit point of } X$$

First, $X \subset \mathbb{R}$ will be bounded if there is a closed interval $I = [a, b]$ such that $X \subset I$.

Now, let's suppose that no limit point p exists, meaning that every point in $\mathbb{R}$ is not a limit point, having either no point or a finite amount of points of X within its neighborhood.

$$\forall p \in \mathbb{R} : U_\delta(p) \cap X \text{ is finite.}$$

$$\text{Let } p \in I \subset \mathbb{R} \rightarrow p \in U_\delta(p)$$

$$p \in I \rightarrow p \in \bigcup_{x \in [a, b]} U_\delta(x)$$

$$\rightarrow I \subset \bigcup_{x \in [a, b]} U_\delta(x)$$

$$\rightarrow \text{The system } \{U_\delta(x)\} \text{ covers } I$$

We built this setup to start from a non-limit point $p \in I$ (conveniently since it could have being any point in $\mathbb{R}$). By definition, the point itself is part of its own neighborhood, and hence part of the union of neighborhoods, reaching the shape of a covering system.

Notice how I (a closed interval) is covered by a system of open intervals $\{U_\delta(x)\}$. With this we can apply Borel-Lebesgue (Lemma 3.22).

$$\exists x_1, \dots, x_n \in I : I \subset U_\delta(x_1) \cup \dots \cup U_\delta(x_n)$$

$$\rightarrow X \subset I \subset \bigcup_{i=1}^n U_\delta(x_i)$$

Intersecting both sides with X :

$$X = X \cap \bigcup_{i=1}^n U_\delta(x_i) = \bigcup_{i=1}^n (X \cap U_\delta(x_i))$$

By construction, each x_i is not a limit point of X , so each set $X \cap U_\delta(x_i)$ is finite. Since X is a finite union (n terms) of finite sets:

$$\Rightarrow X \text{ is finite. } (\rightarrow \leftarrow)$$

□

3.6 Problems & Exercises

Problem 3.1. Show that if I is any system of nested closed intervals, then

$$\sup\{a \in \mathbb{R} \mid [a, b] \in I\} = \alpha \leq \beta = \inf\{b \in \mathbb{R} \mid [a, b] \in I\}$$

and

$$[\alpha, \beta] = \bigcap_{[a, b] \in I} [a, b]$$

Problem 3.2. Let $S = [0, 2] \cap \mathbb{Q}$. Construct a family of open intervals in $\mathbb{R}$ that covers S but has no finite subfamily covering S .

Problem 3.3. Let S be a bounded infinite set of real numbers, and let L be the set of its limit points. Prove that L has a largest element, that is, a limit point $p \in S$ with $q \leq p$ for every limit point q .

Problem 3.4. Let a system of open intervals $\{U_\alpha\}$ cover $[a, b]$. Prove there is a number $\delta > 0$ such that every subinterval of $[a, b]$ of length less than δ lies entirely inside a single interval $\{U_\alpha\}$ of the system.

Problem 3.5. Suppose the points of $[0, 1]$ could be listed as $x_1, x_2, x_3, \dots$ (every point of $[0, 1]$ appears somewhere in the list). Construct a point of $[0, 1]$ that is not in the list, a contradiction.

Problem 3.6. Let $f: [a, b] \rightarrow \mathbb{R}$ be a function such that $f(a) < 0 < f(b)$, and such that for every $x \in [a, b]$ with $f(x) \neq 0$ there exists $\delta > 0$ satisfying

$$\operatorname{sgn} f(y) = \operatorname{sgn} f(x) \quad \text{for all } y \in (x - \delta, x + \delta) \cap [a, b].$$

Prove that there exists $c \in [a, b]$ such that $f(c) = 0$.

A. Proof Techniques Toolkit

I wanted to take the time to review the proof techniques used per chapter. This appendix collects these with an example on each.

A.1 Archimedean Property

Definition: For any complete ordered field F , given any $x \in F$, there exists $n \in \mathbb{N}_F$ such that $n > x$.

Intuition: No matter how large a number you pick, you can always find a natural number bigger than it. This means the natural numbers are unbounded, they grow without limit.

Used to:

- Show sets are bounded (by finding an n that works as an upper bound)
- Show sets are non-empty (by finding a negative n that serves as a lower element)
- Construct rationals or integers around arbitrary real numbers

Example:

Problem: Let $S = \{1 - \frac{1}{n} : n \in \mathbb{N}\}$. Show that S is bounded above.

Solution: By the Archimedean property, for the number 2, there exists $n \in \mathbb{N}$ with $n > 2$. For any element $1 - \frac{1}{n} \in S$, we have

$$1 - \frac{1}{n} < 1 < 2 < n.$$

Hence 2 is an upper bound for S . □

A.2 Density of $\mathbb{Q}$ in $\mathbb{R}$

Definition: Between any two real numbers $\alpha < \beta$, there exists a rational number r such that $\alpha < r < \beta$.

Formally: $\forall \alpha, \beta \in \mathbb{R}$ with $\alpha < \beta$, $\exists r \in \mathbb{Q}$ such that $\alpha < r < \beta$.

Intuition: The rationals are “everywhere” in the reals, no matter how close two real numbers are, you can always squeeze a rational between them. This allows us to:

- Approximate irrational numbers by rational numbers
- Find elements “as close as we want” to a target value
- Bridge constructions between $\mathbb{Q}$ and $\mathbb{R}$

Example:

Problem: Show that between $\sqrt{2}$ and $\sqrt{3}$, there exists a rational number.

Solution: We have $\sqrt{2} \approx 1.414 \dots$ and $\sqrt{3} \approx 1.732 \dots$. Take $r = \frac{3}{2} = 1.5$. Since $1.414 < 1.5 < 1.732$, we have

$$\sqrt{2} < \frac{3}{2} < \sqrt{3},$$

and $\frac{3}{2} \in \mathbb{Q}$. □

A.3 Proof by Contradiction

Definition: To prove a statement P , assume $\neg P$ and derive a contradiction. Since assuming $\neg P$ leads to something impossible, P must be true.

Intuition: “If assuming the opposite leads to nonsense, then the original must be true.” This technique is especially powerful when:

- Direct proof is difficult or unclear
- You want to prove something doesn’t exist
- You want to prove a uniqueness property (by assuming two things exist and deriving a contradiction)

Example:

Problem: Prove that $\sqrt{2}$ is irrational.

Solution: Suppose $\sqrt{2} \in \mathbb{Q}$. Then $\sqrt{2} = \frac{p}{q}$ where $p, q \in \mathbb{Z}$, $q \neq 0$, and $\gcd(p, q) = 1$ (in lowest terms).

Squaring both sides: $2 = \frac{p^2}{q^2}$, so $p^2 = 2q^2$.

Thus p^2 is even, which implies p is even (if p were odd, p^2 would be odd). Write $p = 2k$ for some $k \in \mathbb{Z}$.

Then $(2k)^2 = 2q^2$, so $4k^2 = 2q^2$, hence $q^2 = 2k^2$.

Thus q^2 is even, so q is even.

But then both p and q are even, contradicting $\gcd(p, q) = 1$.

Therefore $\sqrt{2} \notin \mathbb{Q}$. □

A.4 Mathematical Induction

Definition: To prove a statement $P(n)$ holds for all $n \in \mathbb{N}$:

1. *Base case:* Show $P(1)$ is true
2. *Inductive step:* Assume $P(k)$ is true (inductive hypothesis), then prove $P(k + 1)$ is true

Intuition: Like dominoes, if the first one falls (base case) and each falling domino knocks over the next (inductive step), then all dominoes fall. We use induction for:

- Constructing $\mathbb{N}_F$ step-by-step
- Proving properties that hold “for all natural numbers”
- Building sequences or recursive structures

Example:

Problem: Prove that for all $n \in \mathbb{N}$, the sum $1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$.

Solution:

Base case ($n = 1$): LHS = 1, RHS = $\frac{1(2)}{2} = 1$. ✓

Inductive step: Assume the formula holds for $n = k$, i.e.,

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}.$$

We must show it holds for $n = k + 1$:

$$\begin{aligned} 1 + 2 + \cdots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= (k+1) \left[\frac{k}{2} + 1 \right] \\ &= (k+1) \cdot \frac{k+2}{2} \\ &= \frac{(k+1)(k+2)}{2} \end{aligned}$$

This is exactly the formula for $n = k + 1$. By induction, the formula holds for all $n \in \mathbb{N}$. □

A.5 Supremum Arguments

Definition: To prove $s = \sup(S)$, we must show:

1. s is an upper bound: $\forall x \in S, x \leq s$
2. s is the *least* upper bound: $\forall \varepsilon > 0, \exists x \in S$ such that $s - \varepsilon < x \leq s$

Equivalently for (2): For any $s' < s$, there exists $x \in S$ with $x > s'$ (so s' is not an upper bound).

Intuition: The supremum is the “ceiling” of a set. Nothing in the set exceeds it, but you can get arbitrarily close to it from below. This technique is used constantly to:

- Define functions via suprema of rational approximations
- Show that bounds are tight (no smaller bound works)
- Bridge rational approximations to real limits

Example:

Problem: Let $S = \{1 - \frac{1}{n} : n \in \mathbb{N}\}$. Prove that $\sup(S) = 1$.

Solution:

1. *Upper bound:* For any $n \in \mathbb{N}$, we have $\frac{1}{n} > 0$, so $1 - \frac{1}{n} < 1$. Thus 1 is an upper bound for S .
2. *Least upper bound:* Let $\varepsilon > 0$. By the Archimedean property, there exists $N \in \mathbb{N}$ with $N > \frac{1}{\varepsilon}$, so $\frac{1}{N} < \varepsilon$.

Then $1 - \frac{1}{N} \in S$ and

$$1 - \left(1 - \frac{1}{N}\right) = \frac{1}{N} < \varepsilon,$$

meaning $1 - \varepsilon < 1 - \frac{1}{N} \leq 1$.

So within any ε -neighborhood of 1, there exists an element of S . Thus 1 is the least upper bound. □

A.6 Trichotomy in Ordered Fields

Definition: In an ordered field F , for any $x, y \in F$, exactly one of the following holds:

1. $x = y$
2. $x < y$
3. $x > y$

Intuition: Any two elements are either equal or one is strictly larger, no other options exist. This fundamental property is used in:

- Proof by cases (exhaustively considering all three possibilities)
- Proving injectivity (if $f(x) = f(y)$, then x cannot be $<$ or $>$ than y)
- Establishing contradictions in order-preservation arguments

Example:

Problem: Prove that if $x^2 = y^2$ in an ordered field F , then $x = y$ or $x = -y$.

Solution: We have $x^2 - y^2 = 0$, so $(x - y)(x + y) = 0$.

In a field, a product equals zero if and only if at least one factor equals zero. Thus $x - y = 0$ or $x + y = 0$.

- If $x - y = 0$, then $x = y$.
- If $x + y = 0$, then $x = -y$.

By trichotomy, these are the only possibilities (we cannot have both $x - y \neq 0$ and $x + y \neq 0$ simultaneously if $x^2 = y^2$). □